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Towards contextual-based AI: A scoping review of artificial intelligence in X reality for personalized learning

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ABSTRACT

This systematic review synthesizes 54 peer-reviewed studies published between 2019 and 2025 that examine how artificial intelligence (AI) and extended reality (XR) technologies are integrated to support adaptive and personalized learning. The studies were analyzed across multiple dimensions, including learning contexts, AI applications, adaptive input parameters, software and hardware used, and evaluation methods. The findings indicate growing research interest in AI–XR integration, with the majority of studies focused on procedural training and STEM education. Across these studies, AI is frequently used in multifaceted roles, most notably as a provider of real-time adaptive feedback, conversational agent, and a generator of instructional content. Despite these promising developments, the review identifies several critical limitations. While generative AI, particularly large language models (LLMs) such as GPT, has been widely used for conversational interactions, learner profile data remains largely underutilized. Inputs such as prior knowledge and motivation are rarely incorporated. Most implementations rely on a single adaptive strategy, typically driven by performance-based measures such as pre-quiz scores or task completion. As a result, they do not fully exploit the multimodal sensing capabilities of XR platforms (e.g., eye tracking, gesture recognition, environmental tracking), which could support context-sensitive, dynamically generated 3D content aligned with when, where, and how learners need support. Current evaluations of AI–XR systems also remain dominated by short-term performance outcomes, with limited attention to knowledge transfer and critical thinking. These findings highlight key opportunities for designing context-aware, learner-centered AI–XR systems and call for future research that more fully leverages multimodal data, incorporates richer learner profile information, and is grounded in explicit pedagogical models.

1. Introduction

Research on the application of artificial intelligence (AI) in education has been a long-standing area of scholarly interest, gaining formal momentum with the establishment of the International AI in Education (IAIED) Society in 1997 (International Artificial Intelligence in Education Society, 1997; Zawacki-Richter et al., 2019). Since then, advancements in AI technologies have fueled increasing academic attention to their educational potential (Benhaddou et al., 2024; Crompton & Burke, 2023). A systematic review by Zawacki-Richter et al. (2019), which examined educational publications from 2007 to 2018, found a nearly threefold increase in research on AI in education during that period. More recently, the emergence of generative AI, especially large language models (LLMs), has catalyzed a sharp acceleration in scholarly activity (Deng et al., 2025; Ogunleye et al., 2024). As noted by Crompton and Burke (2023), the number of AI-related publications in higher education nearly doubled or tripled between 2021 and 2022 alone.

Among the diverse applications of AI in education, personalized and adaptive learning has emerged as a particularly active area of research (Afini Normadhi et al., 2019; Liu, Guo, Jiao, et al., 2024; Martin et al., 2020; Xie et al., 2019). Educational psychology has long emphasized the importance of accommodating individual differences among learners, highlighting the need to tailor instruction to suit unique needs and preferences (Bernacki et al., 2020; Schreiber et al., 2020). AI presents

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promising solutions to this challenge, with a growing body of literature exploring how learner data can be analyzed to implement adaptive strategies (Li & Wilson, 2025; Sun et al., 2024).

While early AI systems predominantly relied on static rules and a limited set of inputs (e.g., text), recent advances in sensor technologies and immersive platforms have enabled the development of *context-aware AI* applications (Fan et al., 2024; Shui et al., 2024; Vercauteren et al., 2019; Wilson et al., 2025). Immersive technologies with real-time tracking (such as Apple Vision Pro with eye-tracking and environmental sensing) enhance contextual adaptivity for personalized learning (Behravan & Gracanin, 2024; Vaughan et al., 2016; Zahabi & Abdul Razak, 2020). For example, Li et al. developed an adaptive XR tool that integrates conversational scaffolding to support reflective writing. Their work demonstrates that immersive spatial interfaces can enhance engagement, while also presenting usability and health-related challenges that require careful task adaptation (Li, Neshaei, et al., 2025). These systems can leverage real-time environmental data, user behavior, and situational context together to provide timely, relevant, and personalized support for students using adaptive educational technologies (Fan et al., 2024; Shui et al., 2024). Drawing on the framework proposed by Rauschnabel et al. (2022), this review adopts XR ("X Reality") as an umbrella term encompassing virtual reality (VR), augmented reality (AR), and mixed reality (MR), while acknowledging the evolving and interdisciplinary nature of XR terminology.

Despite the promising potential of AI and immersive technologies for context-based learning, relatively few studies have examined their integration within educational contexts. Existing literature often addresses these technologies in isolation, with limited empirical research exploring their combined pedagogical applications (Vaughan et al., 2016; Zahabi & Abdul Razak, 2020). The few studies that do investigate AI within XR environments are typically situated in fields such as computer science or electrical engineering, where the primary emphasis is on technical implementation rather than pedagogical relevance (e.g., Liu, Deng, et al. 2025; Wei et al. 2025). For instance, Vaughan et al. (2016) reviewed autonomous-adaptive training systems in XR across five application areas: medical, industrial, rehabilitation, serious games, and online learning. While the study offers valuable insights into the transferability of XR training to real-world tasks, its primary focus was on technical components such as data input mechanisms, storage systems, and adaptive logic, with limited discussion of instructional strategies or learning outcomes. While some reviews offer valuable perspectives on specific topic areas, they provide limited insight into the broader integration of AI in XR. Hirzle et al. (2023) surveyed interaction design and HCI patterns in XR, but did not synthesize how AI uses multimodal contextual cues (e.g., gaze, gesture, spatial positioning) to support adaptive learning. Zahabi and Abdul Razak (2020) reviewed adaptive XR training systems primarily from a performance and ergonomics perspective, without examining recent advances in AI-driven feedback, learner modeling, or contextual sensing. Xie et al. (2019) focused specifically on VR adaptive learning, without attention to AI or contextual adaptation. As such, questions regarding the role of AI in advancing instructional design, enabling adaptive learning, and supporting pedagogical innovation within educational settings remain underexplored.

To address this gap, the present review synthesizes research on AI-driven adaptive learning within XR environments published between 2019 and 2025, and captures developments following the widespread adoption of generative AI technologies. The review places particular emphasis on educational applications, instructional design, and assessment practices. Following previous efforts (Xie et al., 2019), it is guided by the following research questions:

- **RQ1:** What are the primary applications and functional roles of AI in the design of adaptive learning experiences within XR environments?
- **RQ2:** In which educational contexts is AI applied within immersive XR settings?
- **RQ3:** What types of parameters and input features are leveraged to facilitate adaptive learning through AI in XR-based systems?
- **RQ4:** What types of learning support strategies are employed to facilitate adaptive learning in AI-integrated XR environments?
- **RQ5:** How are learning outcomes and user experiences evaluated in AI-supported immersive XR environments?
- **RQ6:** What hardware platforms and software are employed to implement AI-enhanced XR learning systems?

2. Related work

2.1. AI-enhanced adaptive and personalized learning

The use of artificial intelligence in education has grown rapidly, with particular focus on adaptive and personalized learning. Although these concepts are often conflated, they describe different processes. Personalized learning emphasizes adjustments driven by the learner, where the pace and approach are aligned with individual needs (U.S. Department of Education, Office of Educational Technology, 2017). Adaptive learning, on the other hand, highlights system-driven adjustment. It involves continuous monitoring of learner activity, interpreting behavior through domain models, inferring needs and preferences, updating user models, and dynamically adapting the learning process accordingly (Paramythis & Loidl-Reisinger, 2003). Despite these distinctions, educational research frequently treats the two terms as interchangeable, using both to capture learning experiences shaped by prior knowledge, goals, preferences, and affective states (Raj & Renumol, 2022; Xie et al., 2019; Zhong, 2023). Both approaches share the goal of addressing diverse learning needs, and AI is often recognized for its ability to scale such support across a wide range of contexts (Zhong, 2023).

In the education literature, AI technologies encompass a broad spectrum of computational models, including machine learning (ML) techniques such as decision trees and support vector machines; deep learning (DL) algorithms, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs); rule-based systems that use predefined if-then logic and are commonly employed in intelligent tutoring systems; and LLMs, such as GPT-4, which enable analysis of learners' queries and the generation of customized learning content (Xu & Ouyang, 2022). These technologies can process real-time learner data, model cognitive and affective states, and deliver dynamically tailored content and feedback (Kabudi et al., 2021; Shirazi et al., 2024). Recent studies have explored the application of AI in a range of adaptive and personalized learning strategies, including adjusting instructional difficulty (Bian et al., 2019; Izountar et al., 2021; San Blas et al., 2025; Zheng et al., 2021), modifying instructional content (Gao et al., 2025; Kushwaha & Sharma, 2025; Yannier et al., 2024), generating personalized feedback (Chen et al., 2022; Li, Zhang, et al., 2025; Liu, Guo, Pan, et al., 2024; Yu et al., 2021), and providing scaffolding based on learner performance and progress (Li & Wilson, 2025; Sun et al., 2024). In addition to supporting instructional adaptation, AI systems can automate assessment and deliver instant feedback. They can also predict future learning outcomes. These capabilities collectively position AI as a powerful tool for individualized instruction at scale and for personalized learning (Sarshartehrani & Mohammadrezaei, 2024).

2.2. Intersection of AI and XR for context-aware adaptive learning

In parallel with advances in AI, the emergence of immersive technologies has significantly reshaped the landscape of technology-enhanced learning. In the educational literature, immersive learning environments encompass a wide range of systems, from fully immersive head-mounted displays (e.g., VR headsets with six degrees of freedom) to projection-based systems and mobile AR applications (Hirzle et al., 2023; Lucas-Pérez & Ramírez-Sanz, 2024; Yu, 2023). Given the variability in how VR is defined across disciplines and the rapid evolution of immersive systems, Rauschnabel et al. (2022) introduced the term eXtended Reality (XR) as a unifying concept. XR encompasses the full spectrum of immersive learning experiences, including VR, AR, and MR, and reflects the growing convergence of these technologies.

Unlike traditional 2D learning platforms, XR environments provide a heightened sense of presence and situational relevance factors known to enhance learner motivation and cognitive engagement (Craig et al., 2020; Hirzle et al., 2023). Moreover, XR systems can capture multimodal data through real-time sensing mechanisms such as gaze tracking, motion capture, and biometric feedback. These affordances enable XR to function not only as a medium for immersive learning, but also as a rich source of learner context (Dodd & Antonenko, 2012; Lamb et al., 2018; Rueda et al., 2024). When integrated with AI, therefore, such AI-powered XR systems are capable of interpreting behavioral cues and physiological data to deliver instructional content that adapts to learners' ongoing performance (Elghobashy et al., 2023). For instance, systems can adjust feedback or scaffold tasks in real time, informed by gaze duration, posture, or interaction patterns, thereby enabling adaptive learning in situ. This form of integration brings together the computational adaptivity of AI with the immersive affordances of XR, making possible a new generation of intelligent learning environments (Zwolinski & Kaminska, 2024).

A key concept emerging from this integration is *context-aware AI*: systems that leverage real-time environmental and behavioral data to tailor learning to the learner's specific situation. Abowd et al. (1999) define context as "any information that can be used to characterize the situation of an entity," including users, physical environments, and relevant objects. XR technologies are uniquely suited to support this vision, as they can identify real-world entities the learner is interacting with and deliver just-in-time instructional support. For example, an AI-XR system might detect a student examining a piece of lab equipment and respond by projecting explanatory content directly onto the object or displaying an interactive 3D tutorial. Such applications demonstrate the potential for situated learning, where knowledge is constructed in relation to the physical and social context of the learner (Vercauteren et al., 2019). The convergence of AI and XR thus creates a synergistic framework for adaptive and personalized education. AI contributes advanced modeling and decision-making capabilities, while XR offers the sensory and spatial environment for enacting those decisions in meaningful, embodied ways (Alawneh et al., 2024).

3. Methods

We conducted our scoping review following the PRISMA guidelines (Liberati et al., 2009; Page et al., 2021) and also employed a targeted grey literature search (Paez, 2017) to identify relevant studies. All data used and analyzed are available in the supplementary materials.

3.1. Search strategy and selection procedure

This systematic review followed the PRISMA 2020 guidelines (Page et al., 2021) to ensure replicability. The literature search was conducted across three major academic databases: *Scopus*, *IEEE Xplore*, and the *ACM Digital Library*. The publication date range was limited to 2019–2025 to capture recent developments in the integration of generative AI technologies within educational contexts, a period marked by rapid growth in research on AI-supported learning systems (Jeon et al., 2023; Urban et al., 2024; Xu & Ouyang, 2022).

To reflect this review's focus on adaptive AI in XR environments, search terms included AI-related keywords such as "adaptive AI," "context-aware AI," and "contextual AI," in combination with terms related to immersive learning and personalization, such as "virtual reality," "immersive learning," "personalized learning," and "adaptive learning." We adopted the term "contextual AI" as it is an emerging concept frequently used in recent research to describe AI systems that support personalized learning (Fan et al., 2024; Shui et al., 2024; Vercauteren et al., 2019; Wilson et al., 2025). This terminology also aligns with current research trends in the field. The full list of search terms is provided in Table 1. The initial keyword search yielded a total of 65,610 articles, with 59,828 results retrieved from IEEE Xplore alone. To improve the relevance and alignment of the search results with our research objectives, we applied advanced search filters and venue-specific targeting, following the systematic review methodology recommended by Damarell et al. (2019). We excluded system-oriented venues primarily focused on technical architecture or algorithm development and instead targeted peer-reviewed journal and conference publications in the fields of education, human-computer interaction (HCI), and immersive technologies. Targeted publication venues included conference proceedings such as the *IEEE Conference on Virtual Reality and 3D User Interfaces* and the *IEEE International Symposium on Mixed and Augmented Reality*, as well as journals including the *IEEE Transactions on Visualization and Computer Graphics* and *IEEE Transactions on Multimedia*. After applying these filters, a refined dataset of 5,752 records remained. These articles were screened by title and abstract based on the inclusion and exclusion criteria outlined in Table 2.

Table 1
Search terms used for database and grey literature search.

Topic	Keywords and boolean operators
AI	"adaptive AI" OR "context-aware AI" OR "contextual AI" OR "intelligent tutoring system" OR "embodied AI"
XR	"virtual reality" OR "extended reality" OR "immersive learning"
Personalization/Adaptation	"personalized learning" OR "adaptive learning"

A supplementary grey literature search was also conducted via Google Scholar using the same keyword string applied in the database search. The first ten pages of results were screened, yielding ten additional articles that were incorporated into the review and assessed using the same eligibility criteria. To further enhance the comprehensiveness of the review, we employed both backward and forward snowballing (Wohlin, 2014) on four key articles identified during the initial screening. This process yielded eighteen additional studies that met the inclusion criteria. The complete article selection process, including identification, screening, eligibility assessment, and final inclusion, is illustrated in Fig. 1.

3.2. Inclusion and exclusion criteria

To ensure the relevance of the review, a set of predefined inclusion and exclusion criteria was applied. Studies were included if they (1) involved immersive technologies such as virtual, augmented, or mixed reality; (2) were situated within an educational or training context; (3) employed empirical methods; and (4) examined the integration of AI to support adaptive or personalized learning experiences. Eligible publication types included peer-reviewed journal and conference papers, as well as academically oriented preprints and preliminary works (e.g., workshop papers or demos). Studies were excluded if they (1) were not written in English; (2) focused exclusively on AI or personalized learning without an XR component; (3) addressed XR technologies without incorporating AI or personalization elements; (4) lacked an educational or learning context (e.g., focused solely on healthcare or rehabilitation); or (5) described only system-level development without involving user evaluation. Table 2 provides an overview of these inclusion and exclusion criteria.

3.3. Theoretical framework: constructive learning theory

To structure the conceptual lens of this review, we adopt Constructive Learning Theory, which posits that knowledge is actively constructed by learners through authentic tasks, social interaction, and contextualized experiences (Fan et al., 2024; Piaget, 1970; Xie et al., 2019). Rooted in the work of Piaget (1970) and, Vygotsky (1978), constructivist pedagogy underscores the importance of learners' agency

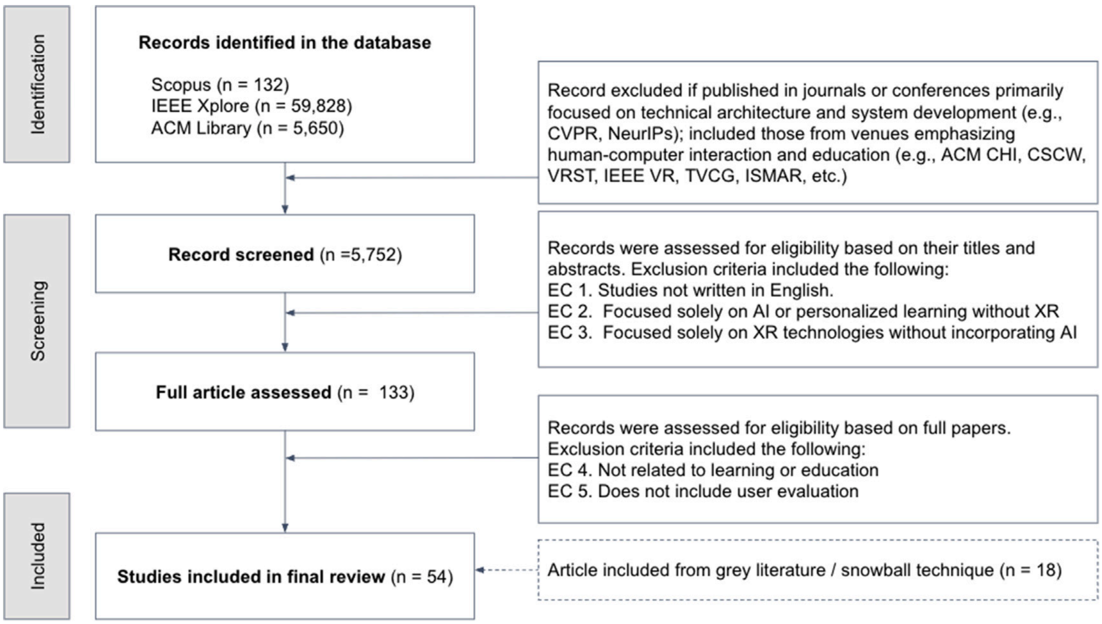


Fig. 1. The PRISMA flowchart.

Table 2
Inclusion and exclusion criteria for study selection.

Inclusion criteria	
IC 1	The study involves immersive technologies, including VR/AR/MR, or immersive environment.
IC 2	The context is a learning environment, such as formal education, training, or simulation-based instruction.
IC 3	The study is empirical in nature (e.g., experimental, quasi-experimental, or qualitative studies with data).
IC 4	The publication includes preprints or non-peer-reviewed yet academically oriented manuscripts.
IC 5	Prototyping or preliminary work is presented (e.g., demo papers, workshop papers, or extended abstracts).
Exclusion criteria	
EC 1	The article is not written in English.
EC 2	The study does not involve XR technologies (VR/AR/MR) or immersive environments, and instead focuses solely on AI or personalized learning.
EC 3	The study does not include AI or personalization elements and focuses only on XR technology without an educational context.
EC 4	The content is not related to learning or education (e.g., focused solely on medical rehabilitation or mental health without an educational component).
EC 5	The study focuses solely on system development and does not include user evaluation.

and the dynamic interplay between prior knowledge and new experiences.

The application of constructivism to technology-enhanced learning environments, especially those involving AI and XR, has been widely recognized in recent educational research. Fu and Hwang (2018), for instance, proposed that effective personalized learning systems should integrate three interconnected components: learning support, system parameters, and learning outcomes (as seen in Fig. 2). These components are embedded in a triadic relationship involving the learner, the environment, and the enabling technology. AI technologies contribute by modeling learners' cognitive and behavioral states and providing personalized feedback, while VR creates immersive environments that contextualize abstract knowledge and provide experiential anchors for meaning-making (Fu & Hwang, 2018; Lucas-Pérez & Ramírez-Sanz, 2024).

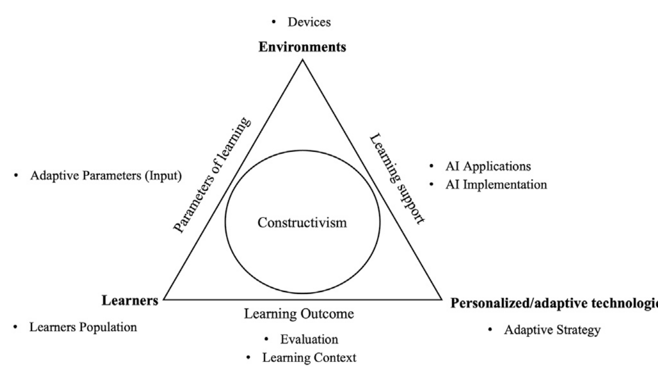


Fig. 2. Theoretical framework adapted from Xie et al. (2019)'s work.

As shown in Fig. 2, each research question in this study aligns with specific components of the framework. RQ1 corresponds to the functional roles of AI within personalized/adaptive technologies; RQ2–RQ5 relate to the learning dimensions of context, parameters, support, and outcomes; and RQ6 corresponds to the technological environment that supports AI-XR implementation. Specifically, *learning support* in this framework refers to how AI systems scaffold the learner's progress through personalized instructional strategies, feedback loops, and adaptive sequencing (Kabudi et al., 2021). *System parameters* represent the underlying computational and pedagogical logic, such as learner modeling and real-time performance metrics, that regulate adaptivity within the system (Coltey et al., 2021; Sarshartehrani & Mohammadrezaei, 2024). Finally, *learning outcomes* encompass not only traditional achievement metrics but also indicators of engagement, attitude, and affect, which are increasingly captured through multimodal data in immersive AI-VR contexts (Elghobashy et al., 2023; Zwolinski & Kaminska, 2024). Based on this framework, we conducted a synthesis of the literature included in this review.

3.4. Data extraction, analysis, and synthesis

To ensure a systematic synthesis of the literature, we developed a structured coding scheme to extract key variables from each study, guided by the framework shown in Fig. 2. Full details of the coding scheme are available in Table A1. The extracted 10 variables were organized across several dimensions, resulting in a total of 74 codes. Author and Year were recorded as the primary citation information to identify and organize the studies.

3.4.1. Learning context

The Learning Context dimension captured the educational domain, learning content, or instructional purpose addressed by each study. The codes were referred to in Conrad et al. (2024) and Jongbloed et al. (2024)'s work. These included procedural training (e.g., driving or surgical tasks), communication skills training (e.g., teacher–student interactions), safety and other applied training, language and writing instruction, general STEM knowledge (e.g., biology, chemistry), declarative knowledge (e.g., historical facts), creative arts (e.g., paper folding), and spatial navigation tasks.

We also coded the Learner Population, including K–12 students, college students, teachers, professionals (such as surgeons and athletes), general audiences, neurodivergent individuals, and other learner groups.

3.4.2. AI applications

To understand how AI supports adaptive and personalized learning, we examined its applications within adaptive learning systems. These applications were coded according to the XR-AI framework proposed by Hirzle et al. (2023), which categorizes AI use in XR environments into five roles: (1) interacting with intelligent virtual agents, (2) using AI to understand users in XR, (3) using AI to create XR worlds, (4) using XR to support AI research, and (5) using AI to interpret user behavior. However, since Hirzle et al. (2023) primarily focuses on system-level implementations rather than educational applications, we also drew upon the framework introduced by Wang, Liu and Adams (2024), which defines AI roles in heuristic learning, including conversational agents, adaptive assessment and feedback, and personalized instruction. Building on these two frameworks, we refined our classification into five categories relevant to educational contexts:

(1) AI as an intelligent conversational agent;
(2) AI for real-time guidance and feedback;
(3) AI for generating instructional content;
(4) AI for detecting and analyzing user gestures or actions; and
(5) AI for inferring learners' cognitive or emotional states.

Additional miscellaneous applications were also noted.

3.4.3. Adaptive learning implementation

We also coded the Adaptive Input Parameters that captured the types of learner data used to drive adaptivity. The term parameters refers to the inputs and variables used in the development of adaptive mechanisms or support systems (Raj & Renumol, 2022). In this review, parameters were coded using the framework proposed by Zahabi and Abdul Razak (2020), which classifies adaptive input variables in XR environments into four main domains: performance-related data, physiological signals, kinematic data, and user profile characteristics. Building on this framework, our analysis identified eight primary types of parameters commonly used to drive adaptive logic in XR-based learning environments:

(1) voice and text inputs,
(2) performance measures,
(3) environmental factors,
(4) gestures and actions,
(5) eye-tracking data,
(6) biometric signals,
(7) user profiles, and
(8) other miscellaneous inputs.

3.4.4. Study method and evaluation

We classified each study based on its Study Methodology, such as mixed methods, quasi-experimental designs, randomized controlled trials, case studies, participatory or action research, other qualitative approaches, or unspecified methods. The Evaluation Metrics dimension captured how learning outcomes and system effects were assessed. These included cognitive or behavioral learning performance, learning efficiency (e.g., time to task completion), usability, engagement with learning or AI, attention and focus, affective responses (e.g., emotions, attitudes), interaction patterns with AI agents (e.g., conversational analysis), perception of AI (e.g., perceived usefulness or trust), and qualitative insights.

3.4.5. Hardware and software implementation

Finally, we referred to Conrad et al. (2024)'s review and recorded the devices used to deliver immersive experiences, including 6DoF head-mounted displays (e.g., Oculus Quest, HTC Vive, HoloLens), 3DoF HMDs (e.g., Google Cardboard), mobile AR platforms, and desktop-based systems. For AI implementation, we documented the underlying technical models used in each study. These included LLMs (e.g., GPT-4, Gemini), traditional machine learning approaches (e.g., SVM, CNN), optimization-based models (e.g., reinforcement learning), Bayesian models, and rule-based systems.

3.4.6. Coding process and inter-rater agreement

Four researchers collaboratively participated in the data extraction process. All four underwent joint training before dividing the workload. Three researchers independently coded distinct subsets of the included studies: the first and second authors each reviewed 10 studies, while the third author coded the remaining studies. The fourth researcher then reviewed all coded data to verify consistency, resolve discrepancies, and finalize the dataset. Interrater reliability was calculated based on this cross-checking process, yielding an agreement rate of 0.94. To ensure consistency in the application of the coding scheme, the research team held calibration meetings during the initial coding phase and met regularly thereafter to address uncertainties and align interpretations. Any ambiguities were resolved through discussion and by jointly reviewing the original articles. The final analyzed dataset can be found in the supplementary material.

4. Results

4.1. Distribution of publications on AI in XR learning environments (2019–2025)

As shown in Fig. 3, the majority of the 54 studies included in this review were published within the past two years, reflecting a pronounced surge in scholarly interest at the intersection of AI and XR in educational contexts. Notably, 77.77 % of the articles (n = 42) were published in 2024 and 2025, with 32 studies appearing between January and October 2025 alone. This notable surge highlights the growing academic interest and the emerging relevance of AI-integrated XR learning environments in recent educational research.

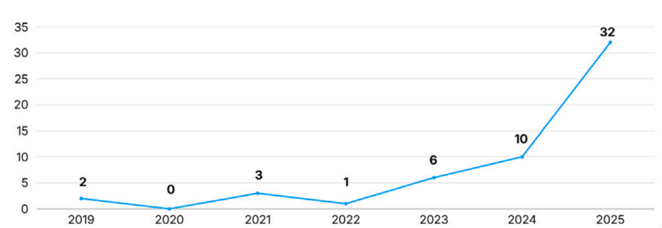


Fig. 3. Distribution of publications on AI in XR learning environments (2019–2025).

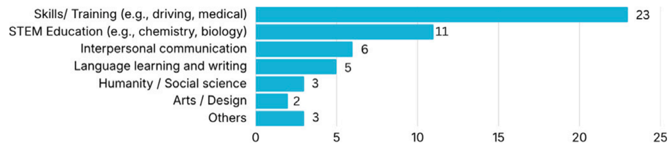


Fig. 4. Distribution of learning topics and educational contents.

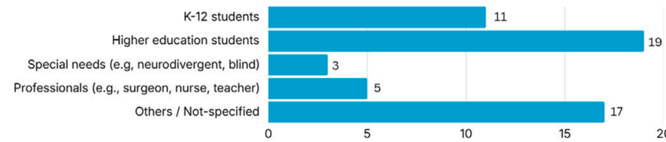


Fig. 5. Distribution of learners.

4.2. Learning contexts: learning topics

Understanding the learning topics addressed by adaptive XR interventions is essential for interpreting their domain-specific learning effectiveness. Different learning domains place distinct cognitive, perceptual, and behavioral demands on learners, and prior research suggests that the affordances of immersive technologies (e.g., embodied interaction, situated practice) may be better suited to some domains than others (Makransky & Petersen, 2021a). Consequently, categorizing studies by their targeted learning topics enables a clearer synthesis of when, how, and for whom adaptive XR produces meaningful learning benefits. As illustrated in Fig. 4, the studies included in this review span a wide range of educational contexts. The most frequently represented category consists of skill-based and training-oriented learning environments (n = 23), where adaptive XR was used to support the acquisition of procedural knowledge and hands-on competencies. This includes applications in surgical and medical training (Bozkir et al., 2019; Daulet et al., 2025; Li, Shao, et al., 2025), industrial assembly and manufacturing (Verstraete et al., 2025), driving simulation (Bian et al., 2019), and sports performance (Liu, Guo, Pan, et al., 2024). The predominance of procedural training aligns with longstanding findings in educational technology research that immersive environments are particularly effective for procedural tasks (Makransky & Petersen, 2021a).

A second major area of application is STEM education (n = 11). For example, these studies use adaptive XR to help students visualize molecular structures in chemistry (Ateş, 2025), understand complex spatial relationships in astronomy (Cheng et al., 2024), receive adaptive support in XR-based computer science learning (Cinar et al., 2024; Gao et al., 2025), and engage with virtual laboratory environments (Xue et al., 2024). Driven by advances in LLMs, adaptive XR has also been applied to interpersonal communication (n = 6), including teacher–student interaction (Zhang et al., 2025) and communication skills training with embodied, conversational AI agents (Yoo et al., 2025). Language learning represents another emerging domain (n = 5) with the use of LLMs. For example, Lee et al. (2023) examined contextual English vocabulary instruction in which XR headsets detect surrounding objects and deliver real-time linguistic scaffolding. In contrast, humanities and social sciences (e.g., Chu et al. 2025) and arts and design (e.g., Chen et al. 2025) remain comparatively underrepresented, each with only a small number of studies. This imbalance suggests that while adaptive XR is gaining substantial traction in technically oriented and performance-based fields, its integration into the humanities and creative disciplines is still in an early exploratory phase.

4.3. Learner distribution

In terms of learner populations (see Fig. 5), higher education students were the most frequently studied group (n = 19), followed closely by K–12 students (n = 11). Studies targeting professionals (e.g., surgeons, nurses, and in-service teachers) accounted for five papers, and a small number of studies focused on special needs learners, including neurodivergent and visually impaired individuals (n = 3). Notably, 17 studies did not clearly specify the learner group. This pattern aligns with prior review studies on personalized online learning, which also reported that higher education and K–12 learners are the most commonly examined populations (Xie et al., 2019).

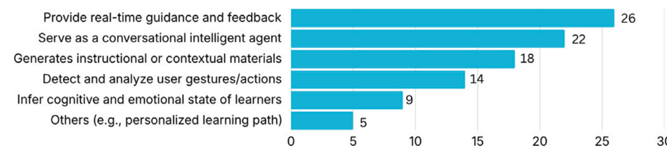


Fig. 6. Applications of AI in learning systems (Note: The counts are non-mutually exclusive due to multiple coding for studies that employed more than one application area.).

4.4. Applications of AI in learning systems

Fig. 6 illustrates the distribution of AI applications and functional roles in the reviewed learning systems. Notably, 28 studies employed AI for two or more functions. The most common application, found in 26 studies, was provision of real-time guidance and feedback. Feedback was typically corrective and delivered through visual, auditory, or text-based modalities during the learning task. For example, Cheng et al. (2024) combined a conversational agent with adaptive feedback, using GPT to provide textual hints when learner responses were incomplete during a science activity. San Blas et al. (2025) designed a system that detected errors in a cycling simulation and offered real-time visual and textual guidance. Many recently published studies implemented AI to detect hand gestures or actions and provided feedback for procedural training (e.g., medical processes) in immersive learning environments (Barenji & Montreuil, 2025; Daulet et al., 2025; Lau et al., 2025; Vannaprathip et al., 2025; Verstraete et al., 2025). Pan et al. (2025) developed a GPT-4 based language tutor in VRChat for users to practice language learning and role play real-life scenarios by chatting in real time. Ateş (2025) developed an ITS-AR system that offered tailored feedback, adaptable learning routes, and targeted assistance to middle school students based on their requirements and progress. The system helped students visualize intricate scientific concepts and experiments using AR technology.

AI also functioned as a conversational agent, as reported in 22 studies. These agents primarily supported learning through chat based, question and answer interactions. For instance, Rychert et al. (2024) implemented a GPT powered assistant to guide problem solving in an escape room scenario. Similarly, Ly et al. (2025) and Chu et al. (2025) used conversational AI to respond to learner inquiries in history museum and science learning contexts, respectively. In another example, Zhang et al. (2025) developed a teacher training simulation in which multiple AI agents portrayed students with distinct personas, enabling preservice teachers to practice classroom interactions. Most studies used visual agents as language or communication tools (Lau et al., 2025; Nguyen et al., 2025; Pan et al., 2025; Yoo et al., 2025). Some studies also used conversational agents within dialogues to support learning or training practices (Vannaprathip et al., 2025).

18 studies used AI to generate instructional content, including quizzes or tasks for assessment/practice (Lau et al., 2025; Liu, Mädche, & Feick, 2025; Ly et al., 2025; Pan et al., 2025), prompts to guide learning activities (Barenji & Montreuil, 2025; Daulet et al., 2025; Lee et al., 2023; Li, Neshaei, et al., 2025; Ren et al., 2025; Shi et al., 2025; Yang et al., 2025a), and narrative stories supporting social-emotional learning for autistic children (Lyu et al., 2024).

Fourteen studies employed AI to detect and analyze user gestures and actions. For example, Zheng et al. (2021) analyzed the toothbrushing behaviors of autistic children using motion and positional data from smart toothbrushes, while San Blas et al. (2025) tracked learner physical behaviors (e.g., steering, braking, and acceleration) during a cycling simulation. AI for user gesture and action detection is also common in procedural training, such as in medical education contexts Barenji and Montreuil (2025); Daulet et al. (2025); Lau et al. (2025); Vannaprathip et al. (2025); Verstraete et al. (2025).

Nine studies focused on inferring learner cognitive and emotional states. Bian et al. (2019) used physiological data to assess engagement among autistic participants in a driving simulation. Similarly, Liu, Mädche, and Feick (2025) analyzed eye-gaze patterns during lectures to estimate learner attention levels.

Six studies explored other applications. Some used AI to analyze surrounding physical environments to enhance contextual learning experiences (Lee et al., 2023; Vaze et al., 2024; Zhang, Chen, et al., 2024a). For language learning and communication, Pan et al. (2025) explored using verbal clues and LLM language abilities, providing verbal encouragement and support that participants appreciated. Daza et al. (2025) conducted detailed analysis of student emotional and cognitive states using facial expressions and other multi-modal data. Yang et al. (2025a) extracted both verbal and nonverbal cues, social factors, and implicit personas, incorporating these social cues into LLM reasoning for social suggestion generation. Others focused on recommending personalized learning paths. For example, Wang, Liu and Adams (2024) and Liu, Guo, Pan, et al. (2024) implemented AI driven recommendation systems for customized learning guidance, while Jaganov et al. (2024), Lau et al. (2025), and Pan et al. (2025) tailored the VR learning environment (e.g., layout, difficulty, and content delivery) based on user preferences and data.

4.5. Parameters and inputs used for adaptive learning implementation

Performance measures were employed in 23 studies (shown in Fig. 7). Metrics included quiz scores (e.g., Wang, Zhang, et al., 2025), task completion rates, and error frequencies (e.g., Bian et al., 2019; Chen et al., 2025; Cinar et al., 2024; San Blas et al., 2025). From studies that measured performance data, the most common adaptive strategies (n = 11) involved adjusting task difficulty (e.g., Bian et al., 2019; Izountar et al., 2021), providing corrective feedback (e.g., Chen et al., 2025; Queisner et al., 2025), or combining both approaches in response to learner outcomes (e.g., San Blas et al., 2025; Zheng et al., 2021). For instance, San Blas et al. (2025) monitored real-time performance by tracking reaction times and compliance with traffic rules in a cycling simulation. These metrics were used to dynamically adjust task complexity and tailor learning scenarios to each learner's evolving needs.

User voice and text inputs were the next most commonly used parameters, appearing in 21 studies. These inputs were primarily leveraged in conversational contexts to interpret learner verbal communication and provide personalized feedback. For instance, Yoo et al. (2025) employed conversational AI agents to support interpersonal skills training and to practice dialogue-based interactions in immersive environments. Gesture and action data were also used in 13 studies. For example, Zheng et al. (2021) analyzed the brushing motions of neurodivergent children to provide personalized instructions for improving technique, while Xue et al. (2024) used hand gestures in a virtual chemistry lab, such as reaching for or manipulating equipment, to trigger context specific instructional support.

Environmental factors were used as adaptive inputs in 12 studies. These included real-world objects and contextual cues from the learner's physical environment, enabling the implementation of context-aware AI systems. For example, Lee et al. (2023) developed an MR language learning application for the HoloLens that delivered instructional content based on the objects detected in the learner's environment. Similarly, Vaze et al. (2024) utilized MR headsets to detect physical objects and deliver relevant content (such as visualizations of plant cell structures) to support situated learning experiences. Eye tracking data were used in eight studies to support gaze based adaptation. For instance, Liu, Mädche and Feick (2025) developed a virtual classroom system that used eye tracking data from the Meta Quest Pro headset to assess learner visual attention to instructional content and deliver adaptive support accordingly.

In addition to these inputs, six studies used learner profile data (e.g., prior knowledge and test scores) to represent relatively static learner characteristics and inform adaptation strategies in XR-based learning systems (Lau et al., 2025; Pan et al., 2025). Four studies incorporated biometric data including heart rate, skin conductance, and muscle activity to drive adaptive responses in XR environments. These biometric signals included electromyography (EMG) and inertial measurement unit (IMU) data (e.g., Liu, Guo, Pan, et al., 2024), heart rate variability (e.g., Wei et al., 2025), functional near-infrared spectroscopy (fNIRS) (e.g., Lingelbach et al., 2023), and respiratory rate and galvanic skin response (e.g., Bian et al., 2019). Lastly, some studies used facial expression data to capture learner emotional states or cognitive load for real-time adaptation (Daza et al., 2025).

4.6. Assessment of learning experiences

As shown in Fig. 8, the most commonly reported assessment type was behavioral or skill-based performance (n = 29), such as task accuracy (Verstraete et al., 2025; Yannier et al., 2024), completion time (e.g., Barenji & Montreuil 2025; Chen et al. 2025; San Blas et al. 2025), or observed proficiency in applied scenarios (e.g., Bian et al. 2019; Daulet et al. 2025; Zhang, Chen, et al. 2024a). This is particularly relevant in training focused domains such as driving, surgery, or sports, where observable performance is central to measuring learning effectiveness. Cognitive performance metrics were also widely used (n = 26), including quiz scores (Chen et al., 2025; Li, Neshaei, et al., 2025; Ly et al., 2025; Wang, Zhang, et al., 2025; Yu et al., 2021), pre/post knowledge tests (Ateş, 2025; Jaganov et al., 2024; Vannaprathip et al., 2025; Yannier et al., 2024), and concept or procedure recall (Daulet et al., 2025; Vaze et al., 2024). These assessments helped researchers quantify learning gains and evaluate the effectiveness of AI/XR enhanced instruction. Measures of engagement and cognitive load were also considered in 19 studies, often through self reports of mental effort or attention, highlighting the cognitive demands of immersive or AI mediated tasks (Ateş, 2025; Barenji & Montreuil, 2025; Bian et al., 2019; Chen et al., 2025; Cheng et al., 2024; Lau et al., 2025; Li, Neshaei, et al., 2025; Ly et al., 2025; Zheng et al., 2021). Many studies incorporated affective or attitudinal measures (n = 18), such as reported student enjoyment, interest, or satisfaction with the learning experience (e.g., Hassan et al.

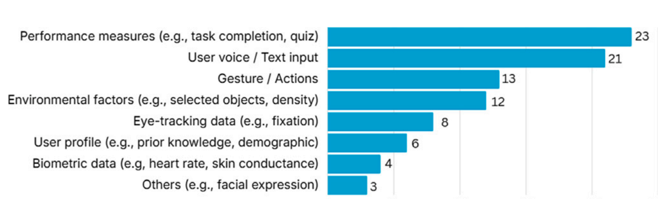


Fig. 7. Parameters/inputs used for adaptive learning implementation (Note: The counts are non-mutually exclusive).

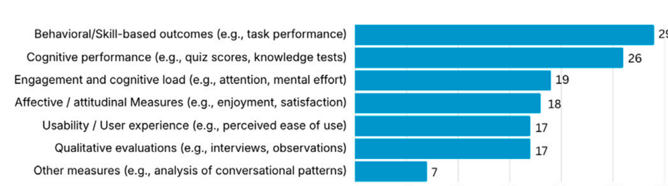


Fig. 8. Assessment of learning experiences (Note: The counts are non-mutually exclusive).

Table 3
Types of AI used.

AI	Literature	Count
GPT models (e.g., OpenAI GPT-3, GPT-4)	(Cheng et al., 2024; Li, Neshaei, et al., 2025; Ly et al., 2025), (Gao et al., 2024; Liu, Mädche, & Feick, 2025; Zhang, Chen, et al., 2024a), (Fuchs et al., 2023; Rychert et al., 2024; Vaze et al., 2024), (Chu et al., 2025; Lee et al., 2023; Lyu et al., 2024), (Gao et al., 2025; Jaganov et al., 2024; Wang, Lin, et al., 2024), (Daza et al., 2025; Shi et al., 2025; Yang et al., 2025a; Yoo et al., 2025), (Lau et al., 2025; Nguyen et al., 2025; Pan et al., 2025)	22
Other LLMs (e.g., LLaMA, Claude)	(Cinar et al., 2024; Wang, Zhang, et al., 2025; Wang, Liu, & Adams, 2024), (Chen et al., 2022; Hassan et al., 2022; Liaw et al., 2023), (Jaganov et al., 2024; Li, Zhang, et al., 2025; Moon & Ke, 2024), (Shi et al., 2025; Yang et al., 2025a)	11
Traditional ML/DL models (e.g., SVM, CNN)	(Lingelbach et al., 2023; Yu et al., 2021; Zhang, Chen, et al., 2024a), (Aloizou et al., 2025; Li, Shao, et al., 2025; Liu, Guo, Pan, et al., 2024), (Bozkir et al., 2019; Lee et al., 2023; Xue et al., 2024), (Fuchs et al., 2023; Izountar et al., 2021; Vaze et al., 2024), (Bian et al., 2019; Daulet et al., 2025; Daza et al., 2025; Ren et al., 2025)	17
Bayesian / Optimization (e.g., Bayesian Knowledge Tracing, BKT)	(Kushwaha & Sharma, 2025; Liu, Guo, Pan, et al., 2024; San Blas et al., 2025; Yannier et al., 2024), (Barenji & Montreuil, 2025; Verstraete et al., 2025)	5
Rule Based	(Bian et al., 2019; San Blas et al., 2025; Zheng et al., 2021), (Harris et al., 2024; Ren et al., 2025)	6
Other / Hybrid models	(Chen et al., 2025; Chu et al., 2025; Lyu et al., 2024), (Fuchs et al., 2023; Queisner et al., 2025; Vaze et al., 2024), (Ateş, 2025; Constantinescu and Iftene, 2025)	8

Table 4
Distribution of XR hardware.

Head-mounted display (HMD)	Count	Non-Head-mounted display (Non-HMD)	Count
Meta Quest (Quest 1, 2, 3, Pro)	14	Projection-Based (e.g., CAVE)	6
HTC Vive (Pro, Focus)	6	Desktop-Based	8
Microsoft HoloLens (1, 2)	4	Mobile AR	4
Others (e.g., Apple Vision Pro, RayNEO, Varjo)	11	Others / Not specified	2

2022; Jaganov et al. 2024; Verstraete et al. 2025). User experience and usability were another important domain (n = 17), capturing perceived ease of use, system intuitiveness, and acceptance (Cinar et al., 2024; Daulet et al., 2025; Li, Zhang, et al., 2025; Liu, Guo, Pan, et al., 2024; Queisner et al., 2025; Yang et al., 2025a). Qualitative evaluations (n = 17), such as interviews and open ended observations, were frequently used to complement quantitative data and provide richer insights into user perceptions and learning processes (Chen et al., 2025; Fuchs et al., 2023; Li, Neshaei, et al., 2025; Liu, Mädche, & Feick, 2025; Ly et al., 2025; Pan et al., 2025; Queisner et al., 2025; Zhang, Chen, et al., 2024a). Less frequently, studies employed other types of measures (n = 7), such as the analysis of conversation patterns, system log data, or interaction traces (Daza et al., 2025; Gao et al., 2024; Moon & Ke, 2024; Zhang et al., 2025), suggesting an emerging interest in using learning analytics and behavioral trace data.

Surveying methodological approaches, 21 studies employed mixed-method designs that combined quantitative assessments with qualitative insights. Others adopted quasi-experimental designs (n = 13), randomized controlled trials (n = 12), or case-based approaches (n = 7). These patterns suggest that the field values both numerical rigor and contextual understanding, with no single methodological paradigm dominating the landscape.

4.7. Hardware and software systems for implementation

As shown in Table 3, the most commonly used software tools in AI-enabled learning experiences were GPT models (n = 22). These were complemented by other LLMs (n = 1), bringing the total number of adaptive learning strategies utilizing LLMs to n = 33. LLMs were frequently deployed as chatbot assistants (e.g., Gao et al. 2024), often enhanced with contextual information (e.g., data about what the user was seeing). In such systems, learners interacted with the LLM by asking questions related to their experience and receiving real-time feedback. LLMs were also used for content generation tasks, such as creating quiz questions, summarizing content, and composing brief instructional texts (e.g. Liu, Mädche, & Feick 2025; Vaze et al. 2024). Traditional machine learning (ML) and deep learning (DL) models were the second most commonly used adaptive logics (n = 17). These models served various purposes, including inferring cognitive and emotional states from facial expressions (e.g. Izountar et al. 2021; Lyu et al. 2024), extracting user intent from text (e.g. Fuchs et al. 2023), enabling speech-to-text and text-to-speech functionality (e.g. Lee et al. 2023), and tracking eye movement data (e.g. Bozkir et al. 2019). Less frequently used were optimization models and Bayesian models (n = 5). Optimization techniques were implemented to support reinforcement learning features (e.g., Kushwaha & Sharma, 2025), while Bayesian models—particularly Bayesian Knowledge Tracing—were employed to estimate learners' conceptual understanding and dynamically adjust task difficulty (e.g., San Blas et al., 2025).

We observed a gradual shift in XR hardware used across the studies in our review. Studies published from 2024 onward increasingly adopted mixed reality experiences that overlay virtual objects onto the physical environment, rather than relying solely on fully immersive virtual reality (Nguyen et al., 2025; Pan et al., 2025; Vaze et al., 2024). This trend tracks the release of next generation XR headsets like the Apple Vision Pro, Meta Quest Pro, and Quest 3, which are equipped with front facing passthrough cameras. Following the recent introduction of lightweight AR smart glasses (e.g., RayNeo AR Glasses), studies such as Yang et al. 2025a employed glasses-based AR for in-situ interaction, further underscoring this trend toward MR/AR-based deployments in educational XR research. Table 4 presents the hardware configurations through which AI driven adaptive learning systems were deployed, broadly grouped into head-mounted displays (HMDs) and non-HMD platforms. HMDs were by far the most common category (n = 35), with the Meta Quest being the most frequently used device (n = 14), followed by HTC Vive (n = 6), HoloLens (n = 4), Apple Vision Pro (n = 1), RayNeo AR Glasses (n = 1), Varjo (n = 3), Oculus Rift S (n = 2), and HP Reverb (n = 1). Among non-HMD platforms, desktop applications were most prevalent (n = 7), followed by projection-based systems (n = 6). Mobile AR applications were the least commonly used configuration (n = 3). Two studies did not report the specific hardware setup used (n = 2).

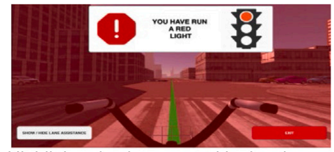
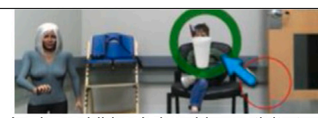
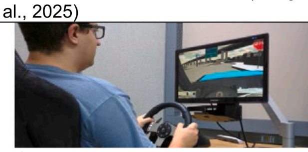
AI application areas along with commonly used parameters and implementation approaches			Adaptive learning strategy example
1. AI to provide real-time guidance and feedback (n = 26)			 Highlight mistake areas with visual guides (Blas et al, 2025); provide verbal guidance (Cheng et al, 2024)
Parameters used	Performance measure	14	
	Gesture / eye-tracking / biometric	13	
	Text/voice inputs	9	
AI implementation	Bayesian / optimization / others	20	
	GPT/other LLMs	11	
	Traditional ML/DL models (e.g., CNN)	5	
2. AI as a conversational and intelligent agent (n = 22)			 Virtual agent answers students' queries in real-time (Li et al., 2025); used in language learning (Pan et al., 2025)
Parameters used	Text / voice inputs	19	
	Performance measure	8	
	Others (e.g., user profile)	5	
AI implementation	GPT	14	
	Other LLMs	7	
3. AI to generate instructional or contextual materials (n = 18)			 Generate learning content (e.g., plant cell) in response to real-world objects (e.g., a plant) detected in the environment. (Vaze et al, 2024)
Parameters used	Performance measure	11	
	Gesture / eye-tracking / biometric	10	
	Text/voice inputs	5	
AI implementation	GPT/other LLMs	16	
	Bayesian / optimization / others	11	
	Others	1	
4. AI to detect and analyze the user gesture / actions (n = 14)			 Analyze children's brushing activity to support toothbrushing instruction (Zheng et al., 2021); understand non-verbal cues in communication (Yang et al., 2025)
Parameters used	Gestures / action	8	
	Eye-tracking / biometrics	3	
AI implementation	Bayesian / optimization / others	12	
	Traditional ML/DL models (e.g., CNN)	5	
	GPT/other LLMs	3	
5. AI to infer cognitive and emotional state (n = 7)			 Infer cognitive load and attention to adjust task difficulty (Bian et al, 2019)
Parameters used	Biometric / eye / facial-tracking	4	
	Text/voice inputs	1	
AI implementation	ML/DL/Bayesian / optimization	6	
	GPT / Other LLMs	3	

Fig. 9. AI application areas along with commonly used parameters and implementation approaches (Note: The counts are non-mutually exclusive due to instances of multiple coding for studies that employed more than one AI implementation, input parameter, or application area.).

4.8. Common implementation practices: how AI is implemented in XR learning environments

Across the reviewed studies, AI was implemented in XR learning environments through a range of functional roles and computational approaches. Building on the typology introduced in Section 4.3, we categorized implementation practices into five primary AI application types commonly observed in XR based learning systems: (1) AI as a conversational agent; (2) AI for providing real-time guidance and feedback; (3) AI for generating real-time instructional materials; (4) AI for detecting and analyzing user gestures and actions; and (5) AI for inferring cognitive and emotional states. Nearly half of the studies (n = 28) employed AI for multiple instructional functions. For example, studies used generative AI both to provide conversational support by responding to student questions after processing their linguistic inputs, and to generate instructional materials informed by student performance data (e.g., task completion, scores) or by specific learning scenarios. (e.g., Lau et al., 2025; Pan et al., 2025).

Fig. 9 summarizes the common input modalities and AI techniques used across these five application categories. Among these, AI-driven real-time guidance and feedback emerged as one of the most prevalent intervention practices (n = 26). Studies in this category drew on a broad set of input parameters, including performance measures (e.g., task completion, error rates, quiz scores; n = 14), gestures, gaze data, and biometric signals (n = 13), and text/voice inputs (n = 9). The nature and modality of feedback also varied widely. For example, San Blas et al. (2025) used visual indicators to highlight errors during a cycling simulation, whereas Cheng et al. (2024) implemented verbal feedback through a mobile AR interface during real-world science activities. These systems most frequently relied on Bayesian, optimization-based, or other rule-driven computational models (n = 20), though a substantial number of studies (n = 11) also incorporated LLMs to generate real-time responses to learners' queries.

When AI was implemented as a conversational agent (n = 22), the predominant method involved processing user voice or text input to enable dialogue-based interactions via virtual agents, typically powered by LLMs such as GPT. While some systems used LLMs solely to answer learner queries (e.g., Chu et al., 2025; Gao et al., 2024), others employed them for more complex tasks. For instance, Wang, Zhang, et al. (2025)

utilized an LLM (i.e., LLaMA) to deliver not only question–answer responses but also contextualized scaffolding and hints during challenging moments in the learning task. Similarly, Pan et al. (2025) designed a GPT-4 based English learning system for social VR that used GPT to assess learner language proficiency, generate role playing scenarios, and sustain conversational practice. This illustrates how LLMs can both facilitate conversational exchanges and provide assessment and dynamic learning content generation for adaptive language learning experiences.

The third category is AI systems designed to generate real-time instructional content (n = 18). In this category, the most commonly used input parameters included learner performance data (e.g., scores, task completion), gestures and biometric signals, and contextual information from the surrounding environment (e.g., object detection). Multimodal input streams enabled systems to dynamically adjust instructional content in response to ongoing learner actions and situational context. Generative AI models, including GPT, other LLMs, and platforms like Wit.AI, were used to create quizzes, explanations, or immersive instructional content in response to learner actions or queries. For example, Vaze et al. (2024) developed a system that identified real world objects (e.g., a plant) and interpreted learner prompts (e.g., "Tell me about this plant") to generate on-demand 3D visualizations of plant cell structures—enabling situated, personalized instruction within MR environments.

In the fourth application category, AI was employed to detect and analyze learners' gestures and actions (n = 14), primarily leveraging motion-based inputs. For example, Zheng et al. (2021) collected data from devices such as Microsoft Kinect and wearable sensors and applied a finite state machine (FSM) to interpret brushing behaviors of neurodivergent children, providing corrective feedback on technique. Across studies in this category, systems typically relied on optimization and rule-based algorithms or conventional machine learning approaches, including computer vision and classification models, to interpret movement patterns and generate real-time analyses.

Finally, studies that used AI to infer cognitive or emotional states (n = 7) relied primarily on biometric and sensor-based data, including heart rate variability, eye-gaze, and facial expression analysis. These inputs were processed through machine learning or deep learning methods to estimate engagement, attention, or cognitive load. For example, Liu, Mädche and Feick (2025) used eye-tracking data from XR headsets to assess visual attention during virtual lectures, informing subsequent instructional adaptation. While promising, this application remains comparatively underexplored, likely due to the technical complexity and ethical considerations associated with collecting and interpreting affective data.

5. Discussion

5.1. Summary of results and research questions

5.1.1. RQ1: application of AI in adaptive XR learning environment

Our findings suggest that AI implementations typically span multiple functions rather than serving a single use case. Several studies (e.g.,) implemented AI systems that simultaneously acted as conversational agents, generated tailored instructional content, and delivered real-time corrective guidance. This finding aligns with existing reviews showing that AI assumes multiple roles (e.g., AI tutors and chatbots, automatic grading and feedback) in educational contexts (Liu, Guo, Jiao, et al., 2024; Zhang, Jaldi, et al., 2024b), and its multifaceted use highlights the flexibility of AI within XR environments.

More than half of the reviewed studies employed AI primarily to provide real-time adaptive feedback during the learning process. Feedback in VR environments is effective in reducing extraneous cognitive load and increasing engagement, with fewer trial-and-error attempts in learning units and better performance in physical, hands-on tasks (Wang et al., 2023). This review extends prior work by examining how AI delivers personalized, real-time support within XR environments. The trend of delivering real-time personalized feedback aligns with the widespread adoption of LLMs (particularly ChatGPT), which can paraphrase hints, generate worked examples, and interpret learner input in natural language (Deng et al., 2025; Ogunleye et al., 2024). The use of AI as a conversational agent or as a generator of instructional materials further reflects the expanding role of generative AI in transforming this area (Chamola et al., 2024).

5.1.2. RQ2: educational contexts and learning domains

XR has been used to target four broad knowledge categories identified in previous studies: procedural, declarative, practical, and perceptual (Conrad et al., 2024; Schloss et al., 2021; Steen et al., 2024). Within this landscape, our review reveals that the vast majority of empirical studies still concentrate on two educational contexts: procedure-based skills training and STEM concept learning. This pattern largely reflects the heavy investment and positive validation of VR for clinical and surgical training in medical education (Jongbloed et al., 2024). Prior research also suggests that this distribution is likely shaped by longstanding funding priorities, the inherent affordances of XR technologies for spatial and procedural tasks, and publication tendencies that favor studies reporting quantifiable performance outcomes (Merchant et al., 2014; Parong & Mayer, 2018; Radianti et al., 2020).

5.1.3. RQ3: parameters and inputs for adaptive learning

Existing reviews on adaptive learning research focus on specific adaptive foci (e.g., personal traits, individual differences, or learning preferences) (Amit et al. 2017; Normadhi et al. 2019). The current review found that a diverse range of parameters is used to drive adaptive learning mechanisms in immersive AI-supported environments, with performance-based measures and language inputs emerging as the most prominent. Task performance such as quiz results, error rates, and response times was the most commonly employed parameter. This reflects a strong tradition of behaviorist assessment logic, in which observable outcomes are directly linked to adaptive strategies (Yusra et al., 2022). As noted earlier, these implementations are primarily limited to domains with well-defined procedural tasks. Such environments offer clear metrics for success, but their applicability to more open-ended or conceptual classroom learning remains limited (Biswas et al., 2021).

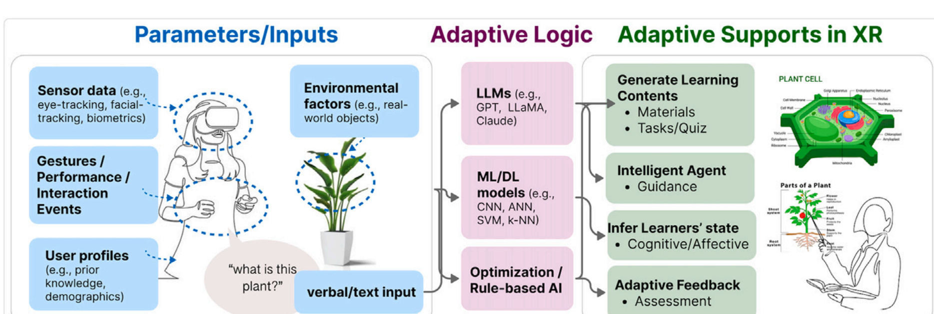


Fig. 10. Contextualized AI in XR for personalized learning: a framework informed by the current scoping review.

Language-based inputs (i.e., text and voice) were also widely used, largely due to the increasing integration of LLMs in XR applications. These models enable adaptive XR systems to interpret questions and interact with learners in real time. As conversational agents supported by generative AI become more embedded in immersive environments, new possibilities for context-aware AI systems emerge, including applications that adapt in response to prior performance, learner intent, confusion, and curiosity. Beyond performance and language, parameters such as environmental factors, eye-tracking, and gesture recognition can be interpreted with the growing capabilities of consumer-grade XR hardware. These inputs hold promise for developing context-sensitive adaptation. Examples of these exciting possibilities include delivering instructional content based on learner focus of attention (Zhang, Chen, et al., 2024a) or interaction with physical objects (Lee et al., 2023; Xue et al., 2024). However, usage remains relatively limited, especially in classroom settings, likely due to space, technical or cost-related constraints (Nemani, 2025).

5.1.4. RQ4: adaptive support strategies

Corrective feedback in multimodal formats was the most common adaptive support strategy. This trend reflects the dominance of performance based adaptation, particularly in training domains where real-time sensor input can identify errors and offer targeted suggestions (e.g., Lee et al. 2023; Vaze et al. 2024; Zheng et al. 2021). In procedural tasks such as surgery, visual and auditory cues were frequently combined, these modalities are especially well-suited for immersive environments, where timely, situated support can enhance motor learning and attention (Lee et al., 2023; Li, Zhang, et al., 2025; Vaze et al., 2024). In addition to feedback, a growing number of systems employed content-level adaptation. The affordances of XR, especially when combined with generative AI, enable just-in-time content generation, as demonstrated in applications ranging from gaze-based scaffolding to environment-aware prompts (Cheng et al., 2024; San Blas et al., 2025).

Conversational assistance represents an emerging but significant form of adaptive support. Unlike feedback that corrects performance, dialogue-based agents offer general guidance, clarification, or encouragement (Gao et al., 2024; Zhang et al., 2025). This form of support is closely tied to the rise of LLMs, which enable naturalistic and context-aware interaction. As seen in studies using AI voice assistants (Gao et al., 2024; Rychert et al., 2024) or AR-integrated chatbots (Cheng et al., 2024; Ly et al., 2025), conversational support may enhance learner autonomy and reduce cognitive load (Gao et al., 2025; Wang, Zhang, et al., 2025).

5.1.5. RQ5: assessment strategies

Behavioral or skill-based assessments are most frequently used in the reviewed studies because metrics such as task accuracy, completion time, and procedural proficiency are well suited and easy to apply to training-oriented domains such as driving, surgery, and sports (Bozkir et al., 2019; Harris et al., 2024; Liu, Guo, Pan, et al., 2024). For example, Harris et al. (2024) found that gaze-adaptive training was successful in generating more expert-like gaze control and putting kinematics, although this did not transfer to improved performance outcomes within the abbreviated training paradigm. These observable but sometimes immeasurable metrics, examined through qualitative analysis, serve as important indicators of learning effectiveness. Cognitive performance assessments such as quizzes, knowledge tests, and recall tasks were also commonly used to quantify learning gains. These objective measures help validate the effectiveness of XR-based instructional designs, particularly when paired with pre/post-test comparisons (Jaganov et al., 2024; Yannier et al., 2024). However, learning in immersive environments is not purely cognitive or behavioral. Affective assessments such as learner interest, satisfaction, and emotional engagement appeared in a substantial portion of studies (e.g., Cheng et al. 2024; Li, Neshaei, et al. 2025). These measures highlight the motivational dimensions of XR learning, especially when AI agents are involved in guiding or encouraging learners (Jonker et al., 2017; Lyu et al., 2024). However, longitudinal studies and assessments remain relatively scarce, pointing to a gap in understanding the long-term impact of AI-supported immersive learning environments (Spantidi & Tracy, 2025).

5.1.6. RQ6: hardware and software systems

Systems reviewed in our study relied heavily on commercial XR devices equipped with multimodal sensors (e.g., Fuchs et al., 2023; Li, Shao, et al., 2025; Xue et al., 2024). An increasing number of recent studies have begun adopting newer headsets, such as the Meta Quest 3, Apple Vision Pro, and RayNeo AR Smart Glasses, which support not only traditional immersive VR but also mixed-reality experiences enabled by integrated pass-through cameras. These devices support real-time eye tracking, gesture recognition, and spatial mapping, all of which are critical for enabling adaptive responses based on user behavior and environmental context. However, technical constraints such as limited processing power, latency issues, and software compatibility, may still hinder the seamless deployment of complex AI models on-device, as noted in previous studies (Wang, Wang, et al., 2025b).

This review also identifies a rapidly evolving technical ecosystem supporting AI-enhanced immersive learning, with a clear trend toward the integration of generative AI models such as ChatGPT and other LLMs (e.g., LLaMA, Gemini). These models have been used to power conversational agents, enable natural language queries, and generate adaptive instructional materials in real time, functions that align well with the demands of learner-centered XR environments (Divekar & Zhou, 2025).

5.2. Implications for research and practice

5.2.1. Current gaps and limitations

Despite promising early developments, our systematic review reveals several gaps in how contextualized AI has been designed and studied within XR-based learning environments. Specifically, we identify six major limitations: (1) the absence of a clear and consistent definition of contextualized AI; (2) an overreliance on single-parameter inputs despite the multimodal sensing capabilities of modern XR devices; (3) a predominant emphasis on performance and skill-based outcomes, with limited attention to knowledge transfer, critical thinking, or metacognitive development; (4) a strong concentration of research in STEM training domains, with little exploration of applications in the humanities and arts; (5) limited consideration of learners' social and cultural contexts; and (6) insufficient engagement with issues related to ethics, privacy, transparency, and accountability.

First, most studies in our review provided little discussion of how AI was theoretically conceptualized or pedagogically positioned within XR learning environments. Although a few studies introduced the terms contextual AI and context-aware AI (e.g., Abowd et al. 1999), these definitions were primarily technical, focusing on contextual computation in which systems adapt to multimodal cues such as location, behavior, and task states (Vercauteren et al., 2019). However, these studies rarely explained how such capabilities were grounded in relevant learning theories, such as contextual learning theory (Hull, 1993), which emphasizes that learning is shaped through interaction with physical, social, and task contexts. Prior reviews echo this concern, noting a general lack of conceptual agreement and coherent definitions of AI and its instructional role within AI-focused education research (Bond et al., 2024; Ng et al., 2021).

Second, although modern XR headsets can capture rich multimodal data streams, including gaze direction, hand motion, head orientation, and physiological indicators, most studies in our review used only a narrow set of inputs. In principle, contextualized AI systems could draw upon these signals to infer learners' affective and cognitive states, adjust task difficulty, or provide individualized feedback (Fan et al., 2024; Shui et al., 2024). In practice, however, the majority of reviewed systems relied on single-parameter inputs. Most commonly, these were performance outcomes such as task completion or pre-post quiz scores (23 studies), or linguistic inputs such as text or voice commands (21 studies). Very few papers integrated environmental tracking, gaze data, or physiological signals, despite the accessibility of such information in current XR hardware. As a result, the potential of multimodal fusion to support meaningful contextualization remains largely unrealized.

Third, research on XR combined with AI overwhelmingly focuses on short-term performance and procedural skill acquisition. As illustrated in Fig. 8 and consistent with prior reviews of XR in education (Huang et al., 2025; Radianti et al., 2020), most studies measured performance based outcomes, such as task accuracy or immediate quiz scores. Although such measures are essential for evaluating training effectiveness, particularly in domains such as medicine and engineering where procedural mastery is a central learning objective, they provide only a partial understanding of learning (Makransky & Petersen, 2021b). None of the studies we reviewed examined whether skills acquired through XR environments with AI support transfer to new contexts or more complex domains nor assessed development in critical thinking, metacognition, or other higher-order outcomes. These omissions are especially concerning in AI-mediated learning, where recent evidence suggests that reliance on AI tools can lead to cognitive offloading and reduce opportunities for critical reasoning (Gerlich, 2025; Jose et al., 2025).

Fourth, the distribution of application domains is heavily skewed toward procedural training and STEM fields, as shown in Fig. 4. This trend can be partly attributed to funding priorities that have historically favored domains such as engineering and medicine over the humanities (Radianti et al., 2020), as well as to the technological affordances of XR interfaces, which align particularly well with procedural training scenarios (Huang et al., 2025). XR is especially well suited for procedural, spatial, and embodied forms of learning, which are central to many STEM disciplines (Makransky & Petersen, 2021b). However, this emphasis has resulted in a limited understanding of how contextualized AI might be used to support learning in the humanities and arts, where XR and AI could foster creativity, interpretive reasoning, historical empathy, and critical media literacy (Bonsu et al., 2025).

Fifth, the reviewed studies rarely considered learners' social or cultural backgrounds when designing adaptive XR and AI systems. Only six papers acknowledged individual differences in learners' profiles, such as prior knowledge or learning preferences. Even in these cases, however, the adaptive mechanisms tended to focus narrowly on performance-related metrics, including prior knowledge and error rates (Lau et al., 2025; Wang, Liu, & Adams, 2024), rather than integrating broader social or cultural dimensions that may shape learners' experiences and needs. This focus contrasts with research demonstrating that culturally grounded and socially responsive design can improve engagement, relevance, and equity in AI-supported learning (Lin et al., 2023; Quishpe-Quishpe et al., 2025).

Lastly, ethical considerations were largely absent from the reviewed literature. None of the studies examined learners' perceptions of transparency, trust, autonomy, or data privacy in XR and AI systems, nor how these factors might influence learning experiences and user perceptions. This pattern echoes findings by Hirzle et al. (2023), who observed that only a small fraction (8 %) of XR and AI research addresses ethical or societal implications.

5.2.2. Directions for future research

The gaps identified above point to several opportunities for future research. First, there is a need to develop clearer, theory-informed frameworks for defining and operationalizing contextualized AI in XR-based learning. Such frameworks should explicitly specify which dimensions of context are considered (e.g., physical, social, cultural, cognitive, affective) and how they relate to targeted learning processes and outcomes. Building on contextual learning theory (Hull, 1993) and prior work on context-aware learning environments, future studies could articulate design principles that distinguish genuinely context-aware systems from more conventional adaptive technologies.

Second, future research could fully exploit the multimodal sensing capabilities of XR devices (e.g., eye tracking, movement trajectories, interaction histories) rather than relying primarily on single performance metrics or language inputs to infer learners' states and tailor scaffolding in real time (Fan et al., 2024). Early deployments already illustrate the potential of such multimodal scaffolding. For example, Vaze et al. (2024) demonstrate how XR environmental tracking can be combined with LLMs to deliver context-sensitive learning content (e.g., dynamically explaining plant cell structures when the learner looks at a plant on the table), and Cheng et al. (2024) show how learners' physical locations and nearby landmarks (e.g., a museum) can be used to create culturally relevant experiences for teaching topics such as climate change. Future work could extend these approaches by systematically comparing unimodal versus multimodal models and by examining trade-offs among data richness, interpretability, and privacy.

Third, future studies could broaden the range of learning outcomes examined in XR + AI research. In addition to immediate performance and usability, researchers could investigate long-term learning and how AI support in XR environments can foster critical thinking and metacognition. Longitudinal and delayed posttest designs would be particularly valuable for assessing whether contextualized AI in XR supports durable knowledge retention rather than only short-term performance gains. Recent experimental work has suggested that MR-based science labs may not always outperform conventional 2D learning on immediate tests, partly due to increased cognitive load, but can support retention through embodied learning (Arjunamahanthi et al., 2025). Given ongoing concerns about cognitive offloading with AI (Gerlich, 2025; Jose et al., 2025), future studies should ground AI–XR scaffolding in evidence-based learning theories that are effective for cultivating critical thinking (e.g., inquiry-based learning) (Pedaste et al., 2015). Experimental designs that compare different forms of AI scaffolding (e.g., explanation prompts aligned with inquiry-based learning versus direct answer provision) could clarify how to support critical engagement through AI and XR.

Fourth, future research could extend contextualized XR + AI beyond STEM and technical training to humanities, arts, and social-emotional domains. Existing work in history education, for example, suggests that XR can foster historical empathy and narrative understanding (Bonsu et al., 2025). Building on this foundation, researchers could explore how contextualized AI might support critical media literacy and creative expression in immersive environments.

Finally, future studies should place greater emphasis on ethics and governance regarding AI and XR, as well as learners' social and cultural contexts. Co-design approaches with diverse learners and educators could help identify which contextual signals are pedagogically meaningful and culturally appropriate, and how AI support should be framed to promote agency and trust (Lin et al., 2023; Quishpe-Quishpe et al., 2025). At the same time, research should examine how different governance models and varying degrees of data transparency affect learners' perceptions of privacy, control, and fairness in XR + AI systems (Hirzle et al., 2023). Such work will be critical for developing contextualized AI that is not only effective, but also responsible and acceptable in real educational settings.

5.2.3. Design and pedagogical implications

Fig. 10 summarizes the parameters, adaptive logic, and learning support strategies used across the reviewed studies, highlighting potential future applications of contextualized AI in XR learning. Overall, the findings point to several practical implications for design and instruction. From a design perspective, developers could adopt a context-first approach to AI integration. Rather than treating AI as a generic conversational layer, designers should identify which contextual cues matter for the targeted learning goals and determine which XR sensing capabilities can capture those cues in a privacy-preserving way. Systems such as

CuriosityXR (Vaze et al., 2024) and the location-aware scaffolds described by Cheng et al. (2024) illustrate how environmental tracking and cultural landmarks can support situated and meaningful learning. Designers could also integrate multimodal feedback strategies that extend beyond text or voice. Gaze-contingent prompts, haptic cues, and adaptive visualizations can draw attention to key features, scaffold complex steps, or prompt reflection (Fan et al., 2024; Scheiter et al., 2019). Transparency and learner control should be treated as core requirements: interfaces should clearly communicate which sensors are active, what data are being used, and how learners can adjust data-sharing settings, which is essential for building trust and supporting informed consent (Hirzle et al., 2023).

Pedagogically, instructors can leverage contextualized AI to provide tailored scaffolding that supports more immersive and targeted learning experiences. Such systems can adapt to individual learners' cognitive and affective states to deliver support that is responsive to real-time needs. For example, Daza et al. (2025) demonstrated how VR headsets can assess learners' cognitive states and attention through facial biometrics and learning metadata, enabling real-time adaptive feedback. Similarly, Moon and Ke (2024) showed how speech mining can be used to infer autistic students' cognitive states and provide personalized scaffolding for social skills training in XR. When AI-generated responses or feedback are incorporated, they can be framed as prompts that encourage learners to monitor, evaluate and reflect on their reasoning, rather than simply providing automated answers, an approach that has been shown to cultivate deeper critical thinking (Jose et al., 2025; Tankelevitch et al., 2024). In adopting XR and AI, it is also crucial to consider the accessibility and logistical challenges associated with XR hardware and data practices, as highlighted in prior literature (Khukalenko et al., 2022). These include physical accessibility constraints of head-mounted displays, variability in students' comfort and tolerance for extended use, and device availability in classroom settings. Addressing these issues is essential to ensure that XR + AI learning experiences are inclusive, feasible, and sustainable in real educational environments.

5.3. Limitation of the present study

This review has several limitations that should be acknowledged. First, the number of studies that met our inclusion criteria was relatively small (n = 54), as we specifically focused on work that examined the use of both AI and XR in educational settings. Although multiple prior reviews have examined AI and XR (e.g., Hirzle et al. 2023; Zahabi & Abdul Razak 2020), these reviews primarily focused on applications outside of education, such as medical training, and often approached the topic from a technical rather than pedagogical perspective. Therefore, the limited number of eligible studies in our review reflects not a sampling issue but the early developmental stage of contextualized AI in XR learning environments. Our grey literature search further indicates that this is an emerging area, with a noticeable surge in publications appearing in 2025 alone (see Supplementary document).

Second, we were unable to provide a detailed synthesis of pedagogical or learning theories underlying the included studies. This limitation stems from the fact that most papers did not articulate a guiding theoretical framework, which made it difficult to analyze how learning theory informed system design or interpretation of outcomes. Nonetheless, our analysis suggests that many AI–XR implementations implicitly align with constructivist and situated learning perspectives, where knowledge construction occurs through meaningful interactions with contextualized environments (Cobb & Bowers, 1999; Hof, 2021).

Third, many papers included in this review did not provide sufficient information about the validity or reliability of their assessment measures. For instance, several studies used gaze metrics, physiological signals, or performance indicators as proxies for constructs such as visual attention or engagement, yet few justified these choices or provided evidence supporting their interpretive validity. In addition, the operationalization of constructs like engagement, attention, and cognitive load varied considerably across disciplines, making comparisons across studies difficult. Due to this inconsistent reporting, it is challenging to determine whether the measures employed in the reviewed studies accurately captured the intended constructs, which limits the confidence with which assessment-related findings can be interpreted. Finally, as with all scoping and systematic reviews, our findings are constrained by the qualitative nature of the analysis. Despite these limitations, this review provides an initial foundation and conceptual framing for understanding how contextualized AI is currently defined, implemented, and studied in XR learning environments.

6. Conclusion

This review synthesized findings from 54 peer-reviewed studies published between 2019 and 2025 that explored how AI and XR technologies are being used to support adaptive and personalized learning. Using a structured 74-code scheme, we examined learning contexts, learner populations, the roles played by AI, adaptive inputs and strategies, assessment practices, and the design of pedagogical user studies. The literature shows clear growth after 2024, reflecting increasing interest in integrating AI-driven adaptivity into immersive learning. With the broader adoption of generative AI and large language models, content creation, real-time feedback, and conversational agents have become common AI functions in XR systems. In terms of learning topics, most studies concentrate on procedural training and STEM subjects, with learners primarily drawn from higher education settings. The reviewed studies frequently report behavioral, cognitive, and affective outcomes, although relatively few investigations track learning over extended periods. The synthesis also highlights several areas where future work can deepen the potential of contextualized AI in XR learning environments. Many implementations focus on a limited set of input signals, even though modern XR devices can capture rich multimodal information such as gaze patterns, motion trajectories, and environmental context. Research has been predominantly centered on STEM and technical training, creating opportunities to extend contextualized AI into humanities, arts, and creative learning domains. Across the reviewed studies, contextual adaptivity was not often tailored to learners' individual profiles or social backgrounds, and learning outcomes tended to focus on short-term effects rather than examining knowledge transfer, critical thinking, or metacognition. Ethical considerations—including privacy, transparency, and governance—are also emerging as important concerns as XR systems become more tightly integrated with AI-driven analytics and multimodal sensing. Together, these findings point to a growing and dynamic landscape for contextualized AI in XR education.

CRediT authorship contribution statement

Zifeng Liu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Serene Cheon:** Methodology, Formal analysis, Data curation. **Austin Stanbury:** Methodology, Formal analysis, Data curation. **Xinyue Jiao:** Writing – review & editing. **Wanli Xing:** Supervision. **Hyo Kang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

See Table A1.

Table A1
The variables and coding scheme used during data extraction. *Note*: Some code categories are not mutually exclusive, and a study may be assigned multiple codes.

Variable	Description and coding scheme
Author	We coded the author(s) of the primary study.
Learning Topics	1. Skills/Training 1.1. Procedural training (e.g., driving, surgical tasks) 1.2. Other applied training (e.g., safety training) 2. STEM Education (e.g., biology, chemistry) 3. Language and Writing 4. Interpersonal Communication (e.g., teacher-student interaction) 5. Social Science, humanity (e.g., historical facts) 6. Creative Arts (e.g., paper folding) 7. Others
Learners	1. K–12 students 2. Higher Education students 3. Professionals (e.g., surgeons, athletes, teachers) 4. Special needs (e.g., Neurodivergent, blind, patients) 5. Others/Not specified
Application/Role of AI	1. Serves as a conversational agent 2. Provides adaptive, real-time feedback 3. Generates instructional or contextual content 4. Detects and analyzes user gestures and actions 5. Infers user's cognitive and emotional states 6. Others (e.g., Analyze surrounding environment, personalized learning path)
Adaptive Parameters/Inputs	1. User Verbal Inputs 1.1. Text input 1.2. Voice input 2. Performance Measures 2.1. Task completion/progress 2.2. Score (e.g., quiz) 3. Environmental factors (e.g., virtual/real world environments, crowd density) 4. Gestures/Actions 5. Eye-tracking data 6. Biometric Data (e.g., Heart rate, Blood pressure, Skin conductance, Muscle activity) 7. User profile (e.g., prior knowledge, demographic) 8. Others (e.g., facial recognition)
Adaptive Strategy	1. Providing Corrective Feedback 1.1. Visual feedback 1.2. Audio feedback 1.3. Text-based feedback 2. Adjusting Task Difficulty 3. Assistance Through Conversational Dialogue (e.g., Answering questions) 4. Generating Adaptive Learning Content and Prompts 5. Others
Study Method	1. Mixed Method Study (Qual + Quant) 2. Quasi-Experimental Design 3. Controlled Experiment (Randomized Controlled) 4. Case study/Small scale user testing 5. Action/participatory design research 6. Other qualitative study 7. Others
Evaluation	1. Cognitive (e.g., quiz score, recall) 2. Behavioral/Skills (e.g., task performance on driving, cycling) 3. Engagement 4. Attention and focus 5. Affective Responses 5.1. Emotional reactions (e.g., satisfaction, anxiety) 5.2. Attitude toward the experience 6. Usability (e.g., ease of use) 7. Others (e.g., conversation pattern with AI agent) 8. Qualitative Findings
Devices	1. 6DoF HMD (e.g., Quest, HTC Vive, Hololens, Vision Pro) 2. 3DoF HMD (e.g., Google Cardboard) 3. Mobile AR 4. Desktop 5. Projected VR (e.g., CAVE)
AI Implementation	1. Large Language Models (LLMs) 1.1. GPT (e.g., GPT-4) 1.2. Other LLMs (e.g., Gemini, Claude, LLaMA, Mistral, Baichuan) 2. Traditional Machine Learning Models (Artificial Neural Networks (ANN), Support Vector Machines (SVM), Convolutional Neural Networks (CNN), Random Forest, k-Nearest Neighbors (k-NN)) 3. Optimization Algorithms (Markov Chain, Reinforcement Learning) 4. Bayesian Models 5. Rule-based AI (e.g., conditional statements) 6. Others

Appendix B. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.caeai.2025.100523.

References

Abowd, G. D., Dey, A. K., Brown, P. J., Davies, N., Smith, M., & Steggles, P. (1999). Towards a better understanding of context and context-awareness. In *Handheld and ubiquitous computing: First international symposium, HUC'99 Karlsruhe, Germany, September 27–29, 1999 proceedings 1* (pp. 304–307). Springer. https://doi.org/10.1007/3-540-48157-5_29

Afini Normadhi, N. B., Shuib, L., Md Nasir, H. N., Bimba, A., Idris, N., & Balakrishnan, V. (2019). Identification of personal traits in adaptive learning environment: Systematic literature review. *Computers & Education*, *130*, 168–190. https://doi.org/10.1016/j.compedu.2018.11.005. https://www.sciencedirect.com/science/article/pii/S0360131518303026.

Alawneh, Y. J. J., Sleema, H., & Salman, F. N. (2024). Adaptive learning systems: Revolutionizing higher education through AI-driven curricula. In *IEEE international conference on advanced learning technologies*. IEEE. https://doi.org/10.1109/ICKECS61492.2024.10616675

Aloizou, V., Linardatou, S., Boloudakis, M., & Retalis, S. (2025). Integrating a movement-based learning platform as core curriculum tool in kindergarten classrooms. *British Journal of Educational Technology*, *56*, 339–365. https://doi.org/10.1111/bjet.13511

Amit, K., Ninni, S., & Jyothi, A. N. (2017). Learning styles based adaptive intelligent tutoring systems: Document analysis of articles published between 2001 and 2016. *International Journal of Cognitive Research in Science, Engineering and Education*, *5*, 83–98.

Arjunamahanthi, P., Kottur, H. R., Stanbury, A., Lam, J. F., Cheon, S., Asadizanjani, N., & Kang, H. (2025). Interactive learning in microelectronics education: Comparing PC and mixed reality approaches for student engagement and visual-spatial memory. In *Proceedings of the extended abstracts of the CHI conference on human factors in computing systems* (pp. 1–8). https://doi.org/10.1145/3706599.3721269

Ateş, H. (2025). Integrating augmented reality into intelligent tutoring systems to enhance science education outcomes. *Education and Information Technologies*, *30*, 4435–4470. https://doi.org/10.1007/s10639-024-12970-y

Barenji, A., & Montreuil, B. (2025). Intelligent tutoring platform for improving psychomotor skills in stem workforce education: XR enhanced itp. *Interactive Learning Environments*, 1–20. https://doi.org/10.1080/10494820.2025.2554998

Behravan, M., & Gracanin, D. (2024). Generative multi-modal artificial intelligence for dynamic real-time context-aware content creation in augmented reality. In *Proceedings of the 30th ACM symposium on virtual reality software and technology* (pp. 1–2). https://doi.org/10.1145/3641825.3689685

Benhaddou, D., El-Hassan, A., Ahmad, K., Ayyash, M., Al-Fuqaha, A. I., Qadir, J., & Iqbal, W. (2024). Data-driven artificial intelligence in education: A comprehensive review. *IEEE Transactions on Learning Technologies*, *17*, 12–31. https://doi.org/10.1109/TLT.2023.3314610

Bernacki, M. L., Hong, W., & Perera, H. N. (2020). A latent profile analysis of undergraduates' achievement motivations and metacognitive behaviors, and their relations to achievement in science. *Journal of Educational Psychology*, *112*, 1409–1430. https://doi.org/10.1037/edu0000445

Bian, D., Wade, J., Swanson, A., Weitlauf, A., Warren, Z., & Sarkar, N. (2019). Design of a physiology-based adaptive virtual reality driving platform for individuals with ASD. *ACM Transactions on Accessible Computing*, *12*, https://doi.org/10.1145/3301498

Biswas, G., Hutchins, N. M., & Zhang, N. (2021). Measuring and analyzing students' strategic learning behaviors in open-ended learning environments. *International Journal of Artificial Intelligence in Education*, *32*, 931–970. https://doi.org/10.1007/s40593-021-00275-x

Bond, M., Khosravi, H., De Laat, M., Bergdahl, N., Negrea, V., Oxley, E., Pham, P., Chong, S. W., & Siemens, G. (2024). A meta systematic review of artificial intelligence in higher education: A call for increased ethics, collaboration, and rigour. *International Journal of Educational Technology in Higher Education*, *21*, 1–45. https://doi.org/10.1186/s41239-023-00436-z

Bonsu, N. O., Zagami, J., Pendergast, D., & Boadu, G. (2025). Virtual reality utilisation in history education: Discovery through a systematic quantitative literature review. *Issues and Trends in Learning Technologies*, *12*, https://doi.org/10.2458/itlt.5995

Bozkir, E., Geisler, D., & Kasneci, E. (2019). Person independent, privacy preserving, and real time assessment of cognitive load using eye tracking in a virtual reality setup. In *2019 IEEE conference on virtual reality and 3D user interfaces (VR)* (pp. 1834–1837). https://doi.org/10.1109/VR.2019.8797758

Chamola, V., Bansal, G., Das, T. K., Hassija, V., Sai, S., Wang, J., Zeadally, S., Hussain, A., Yu, F. R., Guizani, M., & Niyato, D. (2024). Beyond reality: The pivotal role of generative AI in the metaverse. *IEEE Internet Things Mag.*, *7*, 126–135. https://doi.org/10.1109/iotm.001.2300174

Chen, Q., Mishra, R., Purnamasari, L. S., Zanfaly, D., & Kitani, K. (2025). Origami sensei: A mixed reality ai-assistant. In *Proceedings of the 2025 CHI conference on human factors in computing systems*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3706598.3714099

Chen, Y.-L., Hsu, C.-C., Lin, C.-Y., & Hsu, H.-H. (2022). Robot-assisted language learning: Integrating artificial intelligence and virtual reality into English tour guide practice. *Education Sciences*, *12*, 437. https://doi.org/10.3390/educsci12070437

Cheng, A. Y., Guo, M., Ran, M., Ranasaria, A., Sharma, A., Xie, A., Le, K. N., Vinaithirthan, B., Luan, S., Wright, D. T. H., et al. (2024). Scientific and fantastical: Creating immersive, culturally relevant learning experiences with augmented reality and large language models. In *Proceedings of the 2024 CHI conference on human factors in computing systems* (pp. 1–23). https://doi.org/10.1145/3613904.3642041

Chu, X., Wang, M., Spector, J. M., Chen, N.-S., Chai, C. S., Hwang, G.-J., & Zhai, X. (2025). Enhancing the flipped classroom model with generative AI and metaverse technologies: Insights from lag sequential and epistemic network analysis. *Educational Technology Research and Development*, 1–29. https://doi.org/10.1007/s11423-025-10457-2

Cinar, O. E., Rafferty, K., Cutting, D., & Wang, H. (2024). Ai-powered VR for enhanced learning compared to traditional methods. *Electronics*, *13*, 4787. https://doi.org/10.3390/electronics13234787

Cobb, P., & Bowers, J. (1999). Cognitive and situated learning perspectives in theory and practice. *Educational Researcher*, *28*, 4–15.

Coltey, E., Tao, Y., Wang, T., & Vassigh, S. (2021). Generalized structure for adaptable immersive learning environments. In *2021 IEEE 22nd International Conference on Advanced Learning Technologies (ICALT)* (pp. 63–65). IEEE. https://doi.org/10.1109/ICALT52272.2021.00026

Conrad, M., Kablitz, D., & Schumann, S. (2024). Learning effectiveness of immersive virtual reality in education and training: A systematic review of findings. *Computers & Education: X Reality*, *4*, 100053. https://doi.org/10.1016/j.cexr.2024.100053

Constantinescu, G. G., & Iftene, A. (2025). Integrating voice-operated chatbots into virtual reality: A case study on enhancing user interaction. In I. Czarnowski, R. Howlett, & L. Jain (Eds.), *Intelligent decision technologies. KESIDT 2024, volume 411 of smart innovation, systems and technologies*. Singapore: Springer. https://doi.org/10.1007/978-981-97-7419-7_10

Craig, S. D., Roscoe, R. D., Huang, W.-H., & Johnson-Glenberg, M. (2020). Motivation, engagement, and performance across multiple virtual reality sessions and levels of immersion. *Journal of Computer Assisted Learning*, *37*, 745–758. https://doi.org/10.1111/jcal.12520

Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: The state of the field. *International Journal of Educational Technology in Higher Education*, *20*, 1–22. https://doi.org/10.1186/s41239-023-00392-8

Damarell, R. A., May, N., Hammond, S., Sladek, R. M., & Tieman, J. J. (2019). Topic search filters: A systematic scoping review. *Health Information & Libraries Journal*, *36*, 4–40. https://doi.org/10.1111/hir.12244

Daulet, M., Dana, T., & Yevgeniya, D. (2025). Design and evaluation of a context-aware AR system for medical education. *Procedia Computer Science*, *265*, 590–595. https://doi.org/10.1016/j.procs.2025.07.224. https://www.sciencedirect.com/science/article/pii/S1877050925022756. In *20th International Conference on Future Networks and Communications/ 22nd International Conference on Mobile Systems and Pervasive Computing/15th International Conference on Sustainable Energy Information Technology (FNC/MobiSPC/SEIT 2025)*

Daza, R., Shengkai, L., Morales, A., Fierrez, J., & Nagao, K. (2025). Smarte-Vr: Student monitoring and adaptive response technology for e-learning in virtual reality. In *Proceedings of the International workshop on intelligent immersification in the metaverse: AI-driven immersive multimedia* (pp. 15–24). https://doi.org/10.1145/3728487.3759231

Deng, R., Jiang, M., Yu, X., Lu, Y., & Liu, S. (2025). Does Chatgpt enhance student learning? A systematic review and meta-analysis of experimental studies. *Computers & Education*, *227*, 105224. https://doi.org/10.1016/j.compedu.2024.105224. https://www.sciencedirect.com/science/article/pii/S0360131524002380.

Divekar, R., & Zhou, Y. (2025). Immersive, task-based language learning through XR and AI: From design thinking to deployment. *TechTrends*, https://doi.org/10.1007/s11528-025-01048-2

Dodd, B. J., & Antonenko, P. D. (2012). Use of signaling to integrate desktop virtual reality and online learning management systems. *Computers & Education*, *59*, 1099–1108. https://doi.org/10.1016/j.compedu.2012.05.016. https://www.sciencedirect.com/science/article/pii/S0360131512001376.

Elghobashy, Y., Sharaf, N., & Abdennadher, S. (2023). Unleashing the potential: A holistic approach to adaptive learning in virtual reality. In *Emerging technologies and learning* (pp. 44–60). Springer. https://doi.org/10.1007/978-3-031-54327-2_4

Fan, X., Wu, Z., Yu, C., Rao, F., Shi, W., & Tu, T. (2024). Contextcam: Bridging context awareness with creative human-ai image co-creation. In *Proceedings of the CHI conference on human factors in computing systems (CHI '24)* (pp. 1–17). Honolulu, HI, USA: ACM. https://doi.org/10.1145/3613904.3642129

Fu, Q. K., & Hwang, G. J. (2018). Trends in mobile technology-supported collaborative learning: A systematic review of journal publications from 2007 to 2016. *Computers & Education*, *119*, 129–143. https://doi.org/10.1016/j.compedu.2018.01.004

Fuchs, A., Appel, S., & Grimm, P. (2023). Immersive spaces for creativity: Smart working environments. In *2023 international electronics symposium (IES)* (pp. 610–617). IEEE. https://doi.org/10.1109/IES59143.2023.10242458

Gao, H., Huai, H., Yildiz-Degirmenci, S., Bannert, M., & Kasneci, E. (2024). Datalivr: Transformation of data literacy education through virtual reality with chatgpt-powered enhancements. In *2024 IEEE international symposium on mixed and augmented reality (ISMAR)* (pp. 120–129). https://doi.org/10.1109/ISMAR62088.2024.00026

Gao, H., Xie, Y., & Kasneci, E. (2025). Pervrml: Chatgpt-driven personalized VR environments for machine learning education. *International Journal of Human–Computer Interaction*, 1–15. https://doi.org/10.1080/10447318.2025.2504188

Gerlich, M. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. *Societies*, *15*, 6. https://doi.org/10.3390/soc15010006

Harris, D., Donaldson, R., Bray, M., et al. (2024). Attention computing for enhanced visuomotor skill performance: Testing the effectiveness of gaze-adaptive cues in virtual reality golf putting. *Multimedia Tools and Applications*, *83*, 60861–60879. https://doi.org/10.1007/s11042-023-17973-4

Hassan, S. Z., Salehi, P., Røed, R. K., Halvorsen, P., Baugerud, G. A., Johnson, M. S., Lison, P., Riegler, M., Lamb, M. E., Griwodz, C., & Sabet, S. S. (2022). Towards an ai-driven talking avatar in virtual reality for investigative interviews of children. In *Proceedings of the 2nd workshop on games systems* (pp. 9–15). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3534085.3534340

Hirzle, T., Müller, F., Draxler, F., Schmitz, M., Knierim, P., & Hornbæk, K. (2023). When XR and AI meet-a scoping review on extended reality and artificial intelligence. In *Proceedings of the 2023 CHI conference on human factors in computing systems* (pp. 1–45). https://doi.org/10.1145/3544548.3581072

Hof, B. (2021). The Turtle and the mouse: How constructivist learning theory shaped artificial intelligence and educational technology in the 1960s. *History of Education*, *50*, 93–111.

Huang, H.-M., Tai, W.-S., Huang, T.-C., & Lo, C.-Y. (2025). Extended reality in applied sciences education: A systematic review. *Applied Sciences*, *15*, 4038. https://doi.org/10.3390/app15074038

Hull, D. (1993). *Opening minds, opening doors: The rebirth of American education*. Waco, TX: Center for Occupational Research and Development.

International Artificial Intelligence in Education Society. (1997). Founded January 1, 1997; launched with membership from over 40 countries. https://uk.linkedin.com/company/iaied-society (Accessed 2025-Sep-27).

Izountar, Y., Benbelkacem, S., Otmane, S., Khababa, A., Zenati, N., & Masmoudi, M. (2021). Towards an adaptive virtual reality serious game system for motor rehabilitation based on facial emotion recognition. In *2021 international conference on artificial intelligence for cyber security systems and privacy (AI-CSP)* (pp. 1–5). IEEE. https://doi.org/10.1109/AI-CSP52968.2021.9671149

Jaganov, T., Nnadi, C. L., & Watanobe, Y. (2024). The learning labyrinth: Integrating learning theories in VR. In *Proceeding of the 2024 5th Asia service sciences and software engineering conference* (pp. 158–165). https://doi.org/10.1145/3702138.3702156

Jeon, J., Lee, S., & Choe, H. (2023). Beyond Chatgpt: A conceptual framework and systematic review of speech-recognition chatbots for language learning. *Computers & Education*, 104898. https://doi.org/10.1016/j.compedu.2023.104898

Jongbloed, J., Chaker, R., & Lavoué, E. (2024). Immersive procedural training in virtual reality: A systematic literature review. *Computers & Education*, 105124. https://doi.org/10.1016/j.compedu.2024.105124

Jonker, C., Broekens, J., & Moerland, T. (2017). Emotion in reinforcement learning agents and robots: A survey. *Machine Learning*, *107*, 443–480. https://doi.org/10.1007/s10994-017-5666-0

Jose, B., Cherian, J., Verghis, A. M., Varghise, S. M., Mumthas, S., & Joseph, S. (2025). The cognitive paradox of AI in education: Between enhancement and erosion. *Frontiers in Psychology*, *16*, 1550621. https://doi.org/10.3389/fpsyg.2025.1550621

Kabudi, T., Pappas, I., & Olsen, D. H. (2021). Ai-enabled adaptive learning systems: A systematic mapping of the literature. *Smart Learning Environments*, *8*, 1–20. https://doi.org/10.1016/j.caeai.2021.100017

Khukalenko, I. S., Kaplan-Rakowski, R., An, Y., & Iushina, V. D. (2022). Teachers' perceptions of using virtual reality technology in classrooms: A large-scale survey. *Education and Information Technologies*, *27*, 11591–11613.

Kushwaha, R., & Sharma, P. (2025). Enhancing workforce education with VR reinforcement education. In *2025 international conference on intelligent control, computing and communications (IC3)* (pp. 992–998). https://doi.org/10.1109/IC363308.2025.10956200

Lamb, R., Antonenko, P., Etopio, E., & Seccia, A. (2018). Comparison of virtual reality and hands on activities in science education via functional near infrared spectroscopy. *Computers & Education*, *124*, 14–26. https://doi.org/10.1016/j.compedu.2018.05.014. https://www.sciencedirect.com/science/article/pii/S0360131518301143.

Lau, K. H. C., Sen, S., Stark, P., Bozkir, E., & Kasneci, E. (2025). Adaptive gen-ai guidance in virtual reality: A multimodal exploration of engagement in neapolitan pizza-making. In *Proceedings of the 27th international conference on multimodal interaction* (pp. 305–316). https://doi.org/10.1145/3716553.3750760

Lee, H., Hsia, C.-C., Tsoy, A., Choi, S., Hou, H., & Ni, S. (2023). Visionary: Exploratory research on contextual language learning using AR glasses with Chatgpt. In *Proceedings of the 15th biannual conference of the Italian SIGCHI chapter*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3605390.3605400

Li, J., Neshaei, S. P., Müller, L., Rietsche, R., Davis, R. L., & Wambsganss, T. (2025). Spatialearn: Exploring XR learning environments for reflective writing. In *Proceedings of the extended abstracts of the CHI conference on human factors in computing systems* (pp. 1–11). https://doi.org/10.1145/3706599.3719742

Li, M., & Wilson, J. (2025). Ai-integrated scaffolding to enhance agency and creativity in k-12 English language learners: A systematic review. *Information*, *16*, 519. https://doi.org/10.3390/info16070519

Li, W., Shao, J., Shi, P., Li, S., & Yu, H. (2025). Enhancing and shaping closed-loop coadaptive myoelectric interfaces with scenario-guided adaptive incremental learning. *IEEE Journal of Biomedical and Health Informatics*, 1–12. https://doi.org/10.1109/JBHI.2025.3567713

Li, Z., Zhang, H., Peng, C., & Peiris, R. (2025). Exploring large language model-driven agents for environment-aware spatial interactions and conversations in virtual reality role-play scenarios. In *2025 IEEE conference virtual reality and 3d user interfaces (VR)* (pp. 1–11). https://doi.org/10.1109/VR59515.2025.00025

Liaw, S. Y., Tan, J. Z., Lim, S., Zhou, W., Yap, J., Ratan, R., Ooi, S. L., Wong, S. J., Seah, B., & Chua, W. L. (2023). Artificial intelligence in virtual reality simulation for interprofessional communication training: Mixed method study. *Nurse Education Today*, *122*, 105718. https://doi.org/10.1016/j.nedt.2023.105718. https://www.sciencedirect.com/science/article/pii/S0260691723000126.

Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C., Ioannidis, J. P. A., Clarke, M., Devereaux, P. J., Kleijnen, J., & Moher, D. (2009). The Prisma statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: Explanation and elaboration. *Annals of Internal Medicine*, *151*, W–65. https://doi.org/10.1371/journal.pmed.1000100

Lin, X.-F., Wang, Z., & Zhou, W. (2023). Technological support to foster students' artificial intelligence ethics: An augmented reality-based contextualized dilemma discussion approach. *Computers & Education*, *201*, 104813. https://doi.org/10.1016/j.compedu.2023.104813

Lingelbach, K., Diers, D., & Vukelić, M. (2023). Towards user-aware VR learning environments: Combining brain-computer interfaces with virtual reality for mental state decoding. In *Extended abstracts of the 2023 CHI conference on human factors in computing systems*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3544549.3585716

Liu, D., Deng, T., Nam, G., Rong, Y., Pidhorskyi, S., Li, J., Saragih, J., Metaxas, D. N., & Cao, C. (2025). Lucas: Layered universal codec avatars. In *Proceedings of the computer vision and pattern recognition conference* (pp. 21127–21137).

Liu, K.-Y., Guo, T.-Y., Pan, T.-S., Tung, P.-Y., & Lin, Y.-R. (2024). AI batting buddy: A computational and kinematic approach for enhancing batting performance and analysis in baseball. In *Proceedings of the 2024 international conference on multimedia retrieval* (pp. 1246–1250). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3652583.3657590

Liu, S., Mädche, A., & Feick, M. (2025b). Gazeclass: Towards gaze-adaptive cross-device learning support for virtual classrooms. In *Proceedings of the extended abstracts of the CHI conference on human factors in computing systems* (pp. 1–12). https://doi.org/10.1145/3706599.3720232

Liu, Z., Guo, R., Jiao, X., Gao, X., Oh, H., & Xing, W. (2024). How AI assisted k-12 computer science education: A systematic review. In *2024 ASEE annual conference & exposition*. Portland, Oregon. https://doi.org/10.18260/1-2--47532

Lucas-Pérez, G., & Ramírez-Sanz, J. M. (2024). Personalising the training process with adaptive virtual reality: A proposed framework, challenges, and opportunities. In *International conference on extended reality* (pp. 444–456). Springer. https://doi.org/10.1007/978-3-031-71707-9_32

Ly, C., Peng, E., Liu, K., Qin, A., Howe, G., Cheng, A. Y., & Cuadra, A. (2025). Museum in the classroom: Engaging students with augmented reality museum artifacts and generative AI. In *Proceedings of the extended abstracts of the CHI conference on human factors in computing systems* (pp. 1–8). https://doi.org/10.1145/3706599.3719787

Lyu, Y., Liu, D., An, P., Tong, X., Zhang, H., Katsuragawa, K. E. I. K. O., & Zhao, J. I. A. N. (2024). Emooly: Supporting autistic children in collaborative social-emotional learning with caregiver participation through interactive ai-infused and AR activities. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, *8*, 1–36. https://doi.org/10.1145/3699738

Makransky, G., & Petersen, G. B. (2021a). The cognitive affective Model of immersive learning (Camil): A theoretical research-based model of learning in immersive virtual reality. *Educational Psychology Review*, *33*, 937–958.

Makransky, G., & Petersen, G. B. (2021b). The cognitive affective Model of immersive learning (Camil): A theoretical research-based model of learning in immersive virtual reality. *Educational Psychology Review*, *33*, 937–958. https://doi.org/10.1007/s10648-020-09586-2

Martin, F., Chen, Y., Moore, R. L., & Miglani, A. (2020). Systematic review of adaptive learning research designs, context, strategies, and technologies from 2009 to 2018. *Educational Technology Research and Development*, *68*, 1903–1929. https://doi.org/10.1007/s11423-020-09793-2

Merchant, Z., Goetz, E. T., Cifuentes, L., Keeney-Kennicutt, W., & Davis, T. J. (2014). Effectiveness of virtual reality-based instruction on students' learning outcomes in k–12 and higher education: A meta-analysis. *Computers & Education*, *70*, 29–40. https://doi.org/10.1016/j.compedu.2013.07.033

Moon, J., & Ke, F. (2024). Effects of adaptive prompts in virtual reality-based social skills training for children with autism. *Journal of Autism and Developmental Disorders*, *54*, 2826–2846. https://doi.org/10.1007/s10803-023-06021-7

Nemani, S. (2025). Barriers and enablers to adopting virtual reality in lower secondary steam curricula. *Journal of Advanced Research in Education*, *4*, 1–14. https://www.pioneerpublisher.com/jare/article/view/1256.

Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualising AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, *2*, 100041. https://doi.org/10.1016/j.caeai.2021.100041

Nguyen, A., Gul, F., Dang, B., Huynh, L., & Tuunanen, T. (2025). Designing embodied generative artificial intelligence in mixed reality for active learning in higher education. *Innovations in Education and Teaching International*, *62*, 1632–1647. https://doi.org/10.1080/14703297.2025.2499177

Normadhi, N. B. A., Shuib, L., Nasir, H. N. M., Bimba, A., Idris, N., & Balakrishnan, V. (2019). Identification of personal traits in adaptive learning environment: Systematic literature review. *Computers & Education*, *130*, 168–190.

Ogunleye, B., Zakariyyah, K. I., Ajao, O., Olayinka, O., & Sharma, H. (2024). A systematic review of generative AI for teaching and learning practice. *Education Sciences*, *14*, 636. https://doi.org/10.3390/educsci14060636

Paez, A. (2017). Grey literature: An important resource in systematic reviews. *Journal of Evidence-Based Medicine*, *10*, 233–240. https://doi.org/10.1111/jebm.12265

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E.McDonald, S. … Moher, D. (2021). The Prisma 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, *372*, n71. https://doi.org/10.1136/bmj.n71

Pan, M., Kitson, A., Wan, H., & Prpa, M. (2025). Ellma-T: An embodied llm-agent for supporting English language learning in social VR. In *Proceedings of the 2025 ACM designing interactive systems conference* (pp. 576–594). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3715336.3735786

Paramythis, A., & Loidl-Reisinger, S. (2003). Adaptive learning environments and e-learning standards. In *Proceedings of the second European conference on E-learning* (pp. 369–379).

Parong, J., & Mayer, R. E. (2018). Learning science in immersive virtual reality. *Journal of Educational Psychology*, *110*, 785–797. https://doi.org/10.1037/edu0000241

Pedaste, M., Mäeots, M., Siiman, L. A., De Jong, T., Van Riesen, S. A. N., Kamp, E. T., Manoli, C. C., Zacharia, Z. C., & Tsourlidaki, E. (2015). Phases of inquiry-based learning: Definitions and the inquiry cycle. *Educational research review*, *14*, 47–61. https://doi.org/10.1016/j.edurev.2015.02.003

Piaget, J. (1970). *Science of education and the psychology of the child* (D. Coltman, Trans.).

Queisner, M., Remde, C., Luzsa, R., & Sauer, I. M. (2025). Immersive mixed reality training concept for mastering surgical knot-tying. In *2025 IEEE conference on virtual reality and 3d user interfaces abstracts and workshops (VRW)* (pp. 1010–1015). IEEE. https://doi.org/10.1109/VRW66409.2025.00204

Quishpe-Quishpe, L., Acosta-Vargas, I., Rodríguez-Rojas, L., Medina-Arias, J., Coronel-Navarro, D. A., Torres-Gutiérrez, R., & Acosta-Vargas, P. (2025). Impact of a contextualized AI and entrepreneurship-based training program on teacher learning in the Ecuadorian Amazon. *Sustainability*, *17*, 8850. https://doi.org/10.3390/su17198850

Radianti, J., Majchrzak, T. A., Fromm, J., & Wohlgenannt, I. (2020). A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda. *Computers & Education*, *147*, 103778. https://doi.org/10.1016/j.compedu.2019.103778

Raj, N. S., & Renumol, V. G. (2022). A systematic literature review on adaptive content recommenders in personalized learning environments from 2015 to 2020. *Journal of Computers in Education*, *9*, 113–148. https://doi.org/10.1007/s40692-021-00199-4

Rauschnabel, P. A., Felix, R., Hinsch, C., Shahab, H., & Alt, F. (2022). What is XR? Towards a framework for augmented and virtual reality. *Computers in Human behavior*, *133*, 107289. https://doi.org/10.1016/j.chb.2022.107289

Ren, H., Wu, M., Croisdale, G. T., Guo, A., & Wang, X. (2025). Rubikon: Intelligent tutoring for rubik's cube learning through ar-enabled physical task reconfiguration. In *Proceedings of the 2025 ACM designing interactive systems conference* (pp. 3549–3562). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3715336.3735788

Rueda, M. M., Fernández-Batanero, J., & Fernández-Cerero, J. (2024). Possibilities of extended reality in education. *Interactive Learning Environments*, *33*, 208–222. https://doi.org/10.1080/10494820.2024.2342996

Rychert, A., Ganuza, M. L., & Selzer, M. N. (2024). Integrating GPT as an assistant for low-cost virtual reality escape-room games. *IEEE Computer Graphics and Applications*, *44*, 14–25. https://doi.org/10.1109/MCG.2024.3426314

San Blas, H. S., González, S. G., Mendes, A. F. S., González, G. V., & Santana, J. F. D. P. (2025). Improving urban cyclist safety and skills: Integrating a multiagent system and virtual reality training simulations. *Computers and Education Open*, 100255. https://doi.org/10.1016/j.caeo.2025.100255

Sarshartehrani, F., & Mohammadrezaei, E. (2024). Enhancing e-learning experience through embodied AI tutors in immersive virtual environments. In *Human-computer interaction international conference*. https://link.springer.com/chapter/10.1007/978-3-031-60609-0_20.

Scheiter, K., Schubert, C., Schüler, A., Schmidt, H., Zimmermann, G., Wassermann, B., Krebs, M.-C., & Eder, T. (2019). Adaptive multimedia: Using gaze-contingent instructional guidance to provide personalized processing support. *Computers & Education*, *139*, 31–47. https://doi.org/10.1016/j.compedu.2019.05.005

Schloss, K. B., Schoenlein, M. A., Tredinnick, R., Smith, S., Miller, N., Racey, C., & Rokers, B. (2021). The Uw Virtual Brain Project: An immersive approach to teaching functional neuroanatomy. *Translational Issues in Psychological Science*, *7*, 297. https://doi.org/10.1037/tps0000281

Schreiber, J. B., Torsney, B., & Symonds, J. (2020). Silver linings and storm clouds: Divergent profiles of student momentary engagement emerge in response to the same task. *Journal of Educational Psychology*, https://doi.org/10.1037/edu0000605

Shi, J., Jain, R., Chi, S., Doh, H., Chi, H.-G., Quinn, A. J., & Ramani, K. (2025). Caring-ai: Towards authoring context-aware augmented reality instruction through generative artificial intelligence. In *Proceedings of the 2025 CHI conference on human factors in computing systems*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3706598.3713348

Shirazi, B. N., Safavi, A. A., Aftabi, E., & Salimi, G. (2024). The integration of virtual reality and artificial intelligence in educational paradigms. In *2024 11th international and the 17th national conference on E-learning and E-teaching (Icelet)* (pp. 1–6). https://doi.org/10.1109/ICeLeT62507.2024.10493089

Shui, B., Shi, C., Yang, Y., & Nie, X. (2024). Contextvis: Envision contextual learning and interaction with generative models. In *Proceedings of the International conference on human-computer interaction* (pp. 214–224). https://doi.org/10.1007/978-3-031-61953-3_24

Spantidi, O., & Tracy, K. (2025). Impact of gpt-driven teaching assistants in VR learning environments. *IEEE Transactions on Learning Technologies*, *18*, 192–205. https://doi.org/10.1109/TLT.2025.3539179

Steen, C. W., Söderström, K., Stensrud, B., et al. (2024). The effectiveness of virtual reality training on knowledge, skills and attitudes of health care professionals and students in assessing and treating mental health disorders: A systematic review. *BMC Medical Education*, *24*, 480. https://doi.org/10.1186/s12909-024-05423-0

Sun, D., Chou, K. L., Yang, L., & Yang, Y. (2024). A systematic review of technology-supported scaffoldings in empirical studies from 2017-2022. *Educational Technology & Society*, *27*, 185–203. https://www.jstor.org/stable/48787024.

Tankelevitch, L., Kewenig, V., Simkute, A., Scott, A. E., Sarkar, A., Sellen, A., & Rintel, S. (2024). The metacognitive demands and opportunities of generative AI. In *Proceedings of the 2024 CHI conference on human factors in computing systems*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3613904.3642902

Urban, M., Děchtěrenko, F., Lukavský, J., Hrabalová, V., Svacha, F., Brom, C., & Urban, K. (2024). Chatgpt improves creative problem-solving performance in university students: An experimental study. *Computers & Education*, *215*, 105031. https://doi.org/10.1016/j.compedu.2024.105031. https://www.sciencedirect.com/science/article/pii/S0360131524000459.

U.S. Department of Education, Office of Educational Technology. (2017). *Reimagining the role of technology in education: 2017 National Education technology Plan update*. Technical Report. U.S. Department of Education. https://files.eric.ed.gov/fulltext/ED577592.pdf.

Vannaprathip, N., Haddawy, P., Schultheis, H., & Suebnukarn, S. (2025). Sdmentor: A virtual reality-based intelligent tutoring system for surgical decision making in dentistry. *Artificial Intelligence in Medicine*, *162*, 103092. https://doi.org/10.1016/j.artmed.2025.103092. https://www.sciencedirect.com/science/article/pii/S0933365725000272.

Vaughan, N., Gabrys, B., & Dubey, V. N. (2016). An overview of self-adaptive technologies within virtual reality training. *Computer Science Review*, *22*, 65–87. https://doi.org/10.1016/j.cosrev.2016.09.001

Vaze, A., Morris, A., & Clarke, I. (2024). Curiosityxr: Context-aware education experiences with mixed reality and conversation AI. In *2024 IEEE international conference on artificial intelligence and extended and virtual reality (AIxVR)* (pp. 41–49). IEEE. https://doi.org/10.1109/AIxVR59861.2024.00013

Vercauteren, T., Unberath, M., Padoy, N., & Navab, N. (2019). Cai4Cai: The rise of contextual artificial intelligence in computer-assisted interventions. *Proceedings of the IEEE*, *108*, 198–214. https://doi.org/10.1109/JPROC.2019.2946993

Verstraete, A., Geurts, E., & Wijnants, M. (2025). Does one-size training fit all? evaluating adaptive learning for VR assembly training. *Proc. ACM Hum.-Comput. Interact.*, *9*, https://doi.org/10.1145/3734187

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes* (vol. 86). Harvard University Press.

Wang, M., Zhang, D., Zhu, J., & Gu, H. (2025). Effects of incorporating a large language model-based adaptive mechanism into contextual games on students' academic performance, flow experience, cognitive load and behavioral patterns. *Journal of Educational Computing Research*, *63*, 662–694. https://doi.org/10.1177/07356331251321719. original work published 2025.

Wang, W. S., Lin, C. J., Lee, H. Y., Huang, Y. M., & Wu, T. T. (2024). Integrating feedback mechanisms and Chatgpt for vr-based experiential learning: Impacts on reflective thinking and aiot physical hands-on tasks. *Interactive Learning Environments*, *33*, 1770–1787. https://doi.org/10.1080/10494820.2024.2375644

Wang, W.-S., Lin, C.-J., Lee, H.-Y., Wu, T.-T., & Huang, Y.-M. (2023). Feedback mechanism in immersive virtual reality influences physical hands-on task performance and cognitive load. *International Journal of Human–Computer Interaction*, *40*, 4103–4115. https://doi.org/10.1080/10447318.2023.2209837

Wang, X., Liu, S., & Adams, P. (2024). Design and implementation of an ai-enhanced heuristic learning system in a virtual realm. In *Proceedings of the 2024 8th international conference on digital technology in education (ICDTE)* (pp. 255–263). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3696230.3696231

Wang, X., Wang, C., Jia, W., Meng, T., Wang, T., Guo, J., & Tang, Z. (2025b). Empowering edge Intelligence: A comprehensive survey on on-device AI models. *ACM Computing Surveys*, https://doi.org/10.1145/3724420

Wei, J., Siegel, E., Sundaramoorthy, P., Gomes, A., Zhang, S., Vankipuram, M., Smathers, K., Ghosh, S., Horii, H., Bailenson, J., & Ballagas, R. T. (2025). Cognitive load inference using physiological markers in virtual reality. In *2025 IEEE conference virtual reality and 3d user interfaces (VR)* (pp. 759–769). https://doi.org/10.1109/VR59515.2025.00098

Wilson, E., Sendhilnathan, N., Burlingham, C. S., Mansour, Y., Cavin, R., Tetali, S. D., Fernandes, A. S., & Proulx, M. J. (2025). Eye gaze as a signal for conveying user attention in contextual AI systems. In *Proceedings of the 2025 symposium on eye tracking research and applications*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3715669.3727349

Wohlin, C. (2014). Guidelines for snowballing in systematic literature studies and a replication in software engineering. In *Proceedings of the 18th international conference on evaluation and assessment in software engineering*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/2601248.2601268

Xie, H., Chu, H.-C., Hwang, G.-J., & Wang, C.-C. (2019). Trends and development in technology-enhanced adaptive/personalized learning: A systematic review of journal publications from 2007 to 2017. *Computers & Education*, *140*, 103599. https://doi.org/10.1016/j.compedu.2019.103599

Xu, W., & Ouyang, F. (2022). The application of AI technologies in STEM education: A systematic review from 2011 to 2021. *International Journal of STEM Education*, *9*, 1–20. https://doi.org/10.1186/s40594-022-00377-5

Xue, J., Liu, J., Yuan, Q., Yao, Z., Xu, J., & Pan, Z. (2024). Experimental guidance and feedback via operation intention prediction with effect analysis in Chemistry labs. *Education and Information Technologies*, 1–30. https://doi.org/10.1007/s10639-024-12855-0

Yang, B., Guo, Y., Xu, L., Yan, Z., Chen, H., Xing, G., & Jiang, X. (2025). Socialmind: Llm-based proactive AR social assistive system with human-like perception for in-situ live interactions. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, *9*, https://doi.org/10.1145/3712286

Yannier, N., Hudson, S. E., Chang, H., et al. (2024). AI adaptivity in a mixed-reality system improves learning. *International Journal of Artificial Intelligence in Education*, *34*, 1541–1558. https://doi.org/10.1007/s40593-023-00388-5

Yoo, Y., Sun, Y., Shaikh, O., & Won, A. S. (2025). Comparing text-only and virtual reality-embodied conversational AI agents for interpersonal skills training. In *Companion publication of the 2025 Conference on computer-supported cooperative work and social Computing* (pp. 194–198). https://doi.org/10.1145/3715070.3749224

Yu, D. (2023). Ai-empowered Metaverse learning simulation technology application. In *2023 International Conference on intelligent Metaverse Technologies and applications (IMETA)* (pp. 1–6). IEEE. https://doi.org/10.1109/iMETA59369.2023.10294830

Yu, S.-J., Hsueh, Y.-L., Sun, J. C.-Y., & Liu, H.-Z. (2021). Developing an intelligent virtual reality interactive system based on the addie model for learning pour-over coffee brewing. *Computers and Education: Artificial Intelligence*, *2*, 100030. https://doi.org/10.1016/j.caeai.2021.100030. https://www.sciencedirect.com/science/article/pii/S2666920X21000242.

Yusra, A., Neviyarni, S., & Erianjoni, E. (2022). A review of behaviorist learning theory and its impact on the learning process in schools. *International Journal of Educational Dynamics*, *5*, 81–91. https://doi.org/10.24036/ijeds.v5i1.373

Zahabi, M., & Abdul Razak, A. M. (2020). Adaptive virtual reality-based training: A systematic literature review and framework. *Virtual Reality*, *24*, 725–752. https://doi.org/10.1007/s10055-020-00434-w

Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education – where are the educators? *International Journal of Educational Technology in Higher Education*, *16*, 1–27. https://doi.org/10.1186/s41239-019-0171-0

Zhang, C., Chen, T., Shaffer, E., & Soltanaghai, E. (2024). Focusflow: 3d gaze-depth interaction in virtual reality leveraging active visual depth manipulation. In *Proceedings of the 2024 CHI conference on human factors in computing systems* (pp. 1–18). https://doi.org/10.1145/3613904.3642589

Zhang, N., Ke, F., Dai, C.-P., Southerland, S. A., & Yuan, X. (2025). Seeking to support preservice teachers' responsive teaching: Leveraging artificial intelligence-supported virtual simulation. *British Journal of Educational Technology*, *56*, 1148–1169. https://doi.org/10.1111/bjet.13522

Zhang, S., Jaldi, C. D., Schroeder, N. L., López, A. A., Gladstone, J. R., & Heidig, S. (2024). Pedagogical agent design for k-12 education: A systematic review. *Computers & Education*, *223*, 105165. https://doi.org/10.1016/j.compedu.2024.105165. https://www.sciencedirect.com/science/article/pii/S0360131524001799.

Zheng, Z. K., Sarkar, N., Swanson, A., Weitlauf, A., Warren, Z., & Sarkar, N. (2021). Cheerbrush: A novel interactive augmented reality coaching system for toothbrushing skills in children with Autism spectrum disorder. *ACM Transactions on Accessible Computing*, *14*, https://doi.org/10.1145/3481642

Zhong, L. (2023). A systematic review of personalized learning in higher education: Learning content structure, learning materials sequence, and learning readiness support. *Interactive Learning Environments*, *31*, 7053–7073. https://doi.org/10.1080/10494820.2022.2061006

Zwolinski, G., & Kaminska, D. (2024). Flowing through virtual realms: Leveraging artificial intelligence for immersive educational environments. In *Artificial intelligence in education*. Springer. https://doi.org/10.1007/978-3-031-64315-6_4